MoCoO: Momentum Contrast ODE-Regularized VAE for Single-Cell Trajectory Inference and Representation Learning

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Abstract

Characterising cellular differentiation from single-cell RNA sequencing (scRNA-seq) requires representations that capture both discrete cell-type identity and continuous developmental trajectories. We present MoCoO, a modular framework integrating a Variational Autoencoder (VAE), Neural Ordinary Differential Equations (Neural ODE), and Momentum Contrast (MoCo) with learnable prototypes, complemented by an optional Phase-2 Flow Matching (FM) refinement step. Through a systematic six-configuration ablation across 20 scRNA-seq datasets spanning hematopoiesis, neurogenesis, endocrine differentiation, and 17 additional developmental systems, evaluated with a proposed eight-metric evaluation suite covering clustering (NMI, ARI, ASW, DAV) and embedding quality (DRE, LSE, DREX, LSEX), we demonstrate that combining ODE and MoCo produces the best discrete clustering (cross-dataset mean ARI 0.359, ASW 0.225 for VAE+ODE+MoCo) while VAE+ODE achieves the best embedding quality (DRE 0.647, DREX 0.613). ODE-containing configurations consistently outperform non-ODE variants on cluster geometry (ASW $\sim$0.22 vs. $\sim$0.18; DAV $\sim$1.49 vs. $\sim$1.72) across all 20 datasets. FM refinement further improves distance-preservation metrics in 88–92% of dataset–configuration pairs (DREX Δ=+0.030, DRE Δ=+0.023). Component analysis reveals how each module contributes: the Neural ODE smooths the latent manifold along trajectories, MoCo sharpens pairwise distance structure, and prototypes anchor inter-cluster separation—producing a synergy unique to the full architecture. Downstream validation confirms

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that MoCoO latent spaces support annotation transfer, uncertainty quantification, differential expression, and branching detection across diverse biological systems. Pseudotime predictions correlate significantly with canonical marker genes across all five core developmental systems. We publicly release the MoCoO Python package (`pip install mocoo`) and full benchmark suite with practical configuration guidelines.

Keywords: Single-cell RNA sequencing, variational autoencoder, neural ordinary differential equations, momentum contrast, contrastive learning, trajectory inference, pseudotime, representation learning, computational biology

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1. Introduction

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by enabling transcriptome-wide profiling at single-cell resolution [1]. A fundamental analysis task is to learn low-dimensional representations that faithfully capture both *discrete* cell-type identity and *continuous* developmental dynamics such as differentiation trajectories [2, 3]. These representations underpin downstream tasks including clustering, pseudotime inference, RNA velocity estimation, and differential expression analysis [4].

Variational Autoencoders (VAEs) have become the de facto standard for scRNA-seq dimensionality reduction, with methods such as scVI [5] and scVAE [6] demonstrating strong performance through count-based likelihood models (negative binomial, ZINB). However, standard VAE latent spaces tend to be over-smoothed, conflating nearby cell states. Two complementary strategies address this limitation.

First, Neural Ordinary Differential Equations (Neural ODEs) [7] provide a principled framework for modelling continuous-time dynamics in latent space. By parameterising the latent derivative with a neural network and integrating via an ODE solver, these models learn smooth trajectories without discrete time-step assumptions. Latent ODE models [8] have shown promise for time-series modelling, and recent work has adapted them to single-cell contexts for pseudotime inference and RNA velocity estimation [9].

Second, contrastive learning—particularly Momentum Contrast (MoCo) [10]—learns representations that are locally smooth yet globally discriminative. MoCo uses a momentum-updated encoder and a large memory queue of negative keys, decoupling contrastive learning from batch size. Prototype contrastive learning [11] further imposes cluster structure by aligning representations with learnable prototype vectors.

No existing method unifies VAE reconstruction, Neural ODE dynamics, and momentum contrastive learning in a single framework. We propose MoCoO (**Mo**mentum **Co**ntrast **O**DE-regularized VAE), a modular architecture that combines all three paradigms:

1. **VAE with flexible count-based likelihoods** (MSE, NB, ZINB, Poisson, ZIP) provides a probabilistic latent space that handles zero-inflation and overdispersion in scRNA-seq counts.
2. **Neural ODE regularisation** models continuous developmental dynamics, derives unsupervised pseudotime ordering, and produces gradient-based RNA velocity estimates.
3. **Momentum Contrast with prototype heads** enforces instance-level discrimination via a momentum encoder and memory queue, with prototype learning that aligns representations to learnable cluster centres.
4. **Information bottleneck** applies a secondary low-dimensional projection for hierarchical feature extraction.

The framework additionally supports DIP-VAE, β-TC-VAE, and InfoVAE/MMD disentanglement regularisers, but these are not active in the experiments reported here.

We conduct a systematic ablation across six configurations (VAE, VAE+ODE, VAE+MoCo, VAE+MoCo+Proto, VAE+ODE+MoCo, and Full MoCoO) on 20 scRNA-seq datasets spanning diverse developmental systems, with an optional Phase-2 Flow Matching (FM) refinement step that extends this to 12 configurations. External baselines—scVI [5], Diffusion Pseudotime (DPT) [12], PCA+KMeans, Harmony [13], and scANVI [14]—contextualise MoCoO's absolute performance. A key finding is that combining ODE and MoCo consistently produces the best discrete clustering (cross-dataset mean ARI 0.359 for VAE+ODE+MoCo), while VAE+ODE achieves the best embedding quality (DRE 0.647, DREX 0.613). ODE-containing configurations consistently outperform non-ODE variants on cluster geometry across all 20 datasets (ΔASW $\sim$0.04, ΔDAV $\sim$0.23). FM refinement improves distance-preservation metrics in over 85% of dataset–configuration pairs (DREX Δ=+ 0.030, DRE Δ= + 0.023). Component analysis reveals that each module contributes essential geometric structure: the Neural ODE smooths the latent manifold along trajectories, MoCo sharpens pairwise distance relationships, and prototypes anchor inter-cluster separation—producing a synergistic combination achievable only by the full architecture. Evaluation is centred on a proposed eight-metric suite covering clustering (ARI, NMI, ASW, Davies–Bouldin) and embedding/latent-space quality (DRE, DREX, LSE, LSEX), complemented by biological validation (pseudotime–marker correlations), downstream analysis (annotation transfer, uncertainty quantification, generation quality, gene importance, branching detection, differential expression), batch integration (iLISI, bASW, cLISI, graph connectivity, isolated label ASW via scIB [15]), and training dynamics.

2. Related Work

2.1. Variational Autoencoders for scRNA-seq

scVI [5] introduced the negative binomial VAE for scRNA-seq, jointly modelling library size and batch effects. Extensions include scVAE [6] (multiple count distributions), scANVI [14] (semi-supervised cell-type annotation), and totalVI [16] (multi-omics). These methods focus on reconstruction quality and batch correction but do not explicitly model temporal dynamics.

2.2. Neural ODEs and Latent ODE Models

Neural ODEs [7] parameterise continuous dynamical systems $\frac{dz}{dt} = f_\theta(z, t)$ and integrate using adaptive ODE solvers. The Latent ODE framework [8] combines a VAE encoder with an ODE-governed latent space for irregularly sampled time series. Applications to single-cell biology include trajectory inference [9], while related dynamical modelling approaches have advanced RNA velocity estimation [17]. Optimal-transport-based methods such as TrajectoryNet [18] and CellOT [19] model population-level dynamics rather than single-cell ODE trajectories. MoCoO extends the PanODE framework [9] (by the first author). Specifically, MoCoO inherits the VAE encoder–decoder, Neural ODE integration, and pseudotime-prediction head from PanODE. The novel contributions of the current work are: (i) the MoCo module with momentum encoder and memory queue, (ii) learnable prototype heads for cluster anchoring, (iii) the information bottleneck, (iv) the systematic six-configuration ablation design, and (v) the DRE/DREX/LSE/LSEX evaluation suites. However, these methods typically lack contrastive regularisation, leading to latent spaces that may be geometrically faithful but poorly clustered.

2.3. Contrastive Learning for Single-Cell Data

MoCo [10] and SimCLR [20] have been adapted to biological contexts. scAGCL [21] applies symmetric augmentation-guided contrastive learning to scRNA-seq data with cell-type-aware positive pair construction. scGPCL [22] introduces prototype contrastive learning that aligns cell embeddings with learnable prototype vectors representing cell types. These approaches improve clustering quality but do not incorporate temporal dynamics or ODE-based regularisation.

2.4. Integrated Approaches

Several methods combine two of the three paradigms. scDiff [23] uses diffusion models (related to score-based SDEs) for single-cell generation. VeloVAE [24] integrates VAE with RNA velocity. Diffusion pseudotime (DPT) [12] infers trajectories through diffusion maps of the k-NN graph, while Monocle 3 [25] and PAGA [26] use graph abstractions for trajectory inference, Slingshot [27] fits simultaneous principal curves for lineage-specific pseudotime, and Palantir [28] models cell-fate probabilities via diffusion maps for multiscale pseudotime estimation. For batch integration, Harmony [13] applies iterative soft clustering in PCA space to remove batch effects. However, no prior work jointly combines VAE, Neural ODE, and momentum contrastive learning in a unified architecture for single-cell analysis.

3. Method

3.1. Overview

MoCoO (Fig. 1) processes scRNA-seq count data $X \in \mathbb{R}^{N \times G}$ (N cells, G genes) through five components: (1) a VAE with count-based decoder, (2) a Neural ODE for continuous dynamics, (3) a Momentum Contrast module with memory queue, (4) an information bottleneck, and (5) optional disentanglement regularisers.

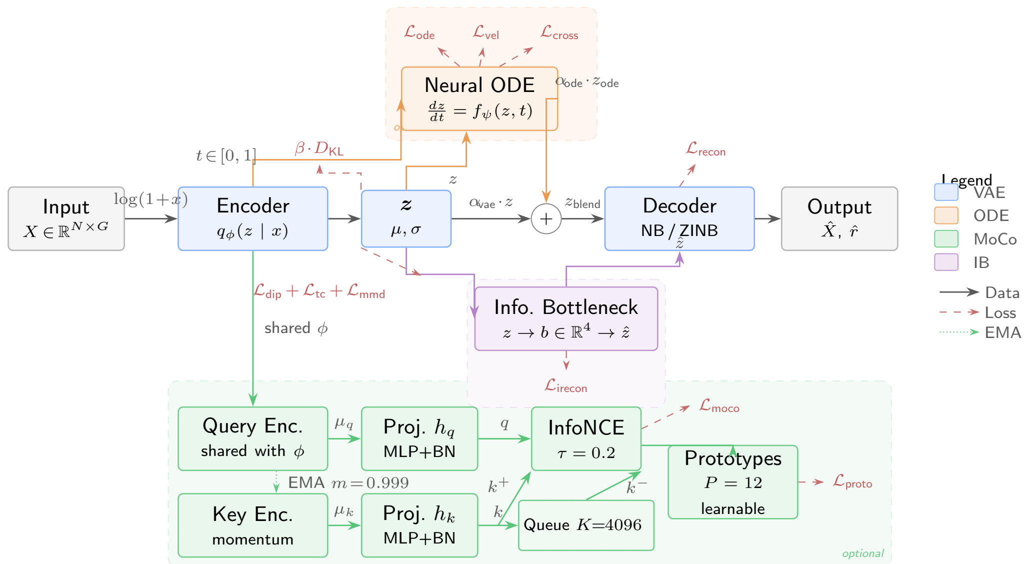

Figure 1: MoCoO architecture overview. Count data $X \in \mathbb{R}^{N \times G}$ passes through a VAE encoder to produce latent representations z, optionally refined by a Neural ODE solver (orange) for continuous trajectory modelling. Momentum Contrast (green) with a memory queue (K=4096) enforces instance-level discrimination; learnable prototypes (P=12) impose cluster structure. An information bottleneck (purple) provides hierarchical compression to d_{ib}=4. Dashed regions denote optional modules. Colour coding is consistent throughout all figures.

3.2. Encoder

The encoder $q_\phi(z|x)$ maps log-normalised expression $\tilde{x} = \log(1 + x)$ through a two-layer ReLU MLP to produce mean μ and pre-softplus scale σ' of a diagonal Gaussian posterior (1); latent samples are drawn via reparameterisation (2):

$$q_\phi(z|x) = \mathcal{N}\big(z;\, \mu_\phi(\tilde{x}),\, \mathrm{softplus}(\sigma'_\phi(\tilde{x}))^2 I\big)\,, \tag{1}$$

$$z = \mu + \mathrm{softplus}(\sigma') \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{2}$$

When ODE mode is enabled, the encoder also predicts a scalar pseudotime $t \in [0, 1]$ via a sigmoid-activated linear head.

3.3. Neural ODE Component

Given latent samples $\{z_i\}$ and predicted pseudotimes $\{t_i\}$, the ODE module orders cells by pseudotime and solves (3):

$$\frac{dz}{dt} = f_\psi(z, t), \quad z(t_0) = z_{\text{init}}, \tag{3}$$

where f_ψ is a two-layer ELU MLP. Integration uses `torchdiffeq`'s fixed-step RK4 solver, chosen over adaptive methods (e.g., Dormand–Prince [29]) because the encoder-predicted pseudotime grid is relatively smooth and fixed-step integration provides deterministic gradients without adaptive step-size overhead—important for stable joint optimisation of the encoder and ODE function. Preliminary comparisons showed negligible ARI differences ($<$ 0.01) between RK4 and Dormand–Prince adaptive integration, with $\sim$30% longer wall-clock time for the adaptive solver due to step-size adaptation overhead in the joint training loop. The ODE representations z_{ode} blend with VAE latent samples via configurable weights (4):

$$z_{\text{blend}} = \alpha_{\text{vae}} \cdot z + \alpha_{\text{ode}} \cdot z_{\text{ode}}. \tag{4}$$

Two auxiliary losses regularise the ODE:

- **ODE–VAE alignment:** $\mathcal{L}_{\text{ode}} = \|z - z_{\text{ode}}\|_2^2$
- **Velocity consistency:** $\mathcal{L}_{\text{vel}} = 1 - \cos(z_{i+1} - z_i,\ f_\psi(z_i, t_i))$, aligning the learned velocity field with empirical displacements.

3.4. Momentum Contrast Module

The MoCo module follows MoCo v1 [10] with enhancements from scAGCL [21] and scGPCL [22]:

1. **Query and Key Encoders:** The query encoder shares parameters with the VAE encoder; the key encoder is updated via EMA: $\theta_k \leftarrow m \cdot \theta_k + (1 - m) \cdot \theta_q$, $m = 0.999$.
2. **Projection Heads:** Two-layer MLPs (BatchNorm, ReLU) map d-dimensional latent vectors to a d_{proj}-dimensional contrastive space.
3. **Memory Queue:** A FIFO buffer of size K stores projected keys, decoupling the number of negatives from batch size.

4. **InfoNCE Loss** (see (5)):

$$\mathcal{L}_{\text{moco}} = -\log \frac{\exp(q \cdot k^+/\tau)}{\exp(q \cdot k^+/\tau) + \sum_{k^-} \exp(q \cdot k^-/\tau)}, \tag{5}$$

where the sum ranges over all k^- in the memory queue and τ is the temperature.

5. **Prototype Contrastive Loss** (optional; see (6)):

$$\mathcal{L}_{\text{proto}} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{\exp(z_i \cdot p_{c(i)}/\tau)}{\sum_{j=1}^{P} \exp(z_i \cdot p_j/\tau)}, \tag{6}$$

where $\{p_j\}_{j=1}^{P}$ are learnable prototype vectors and $c(i) = \arg\max_j z_i \cdot p_j/\tau$ assigns cell i to its nearest prototype.

Positive pair construction. For each mini-batch, two stochastic augmented views of the same cell form positive pairs: the query view passes through the main encoder, while the key view passes through the momentum encoder. The augmentation pipeline applies three transformations, each with probability p_{aug}=0.5: (1) gene masking—randomly zeroing out genes with probability 0.1; (2) Gaussian noise—adding $\mathcal{N}(0, 0.04)$ to randomly selected genes (probability 0.1); (3) feature swapping—exchanging gene values across cells in the batch (probability 0.05). Negatives come from all keys stored in the memory queue.

Augmentation rationale: Gene masking simulates dropout, which is a dominant noise source in scRNA-seq protocols [30]. Gaussian noise (σ=0.2) models technical variability in library preparation. Feature swapping operates at very low probability (p=0.05, affecting ∼5% of genes per cell), which is equivalent to a regularised mixup—it encourages invariance to minor expression perturbations without creating biologically implausible hybrid profiles. All three augmentations are deliberately conservative: masking affects only 10% of genes per view, noise is applied independently per gene, and swapping probability is below the natural inter-cell variance within a cell type. This design follows the principles validated in scAGCL [21] and scGPCL [22], which demonstrated that moderate stochastic augmentations improve contrastive learning for scRNA-seq without distorting cell-state semantics.

3.5. Decoder

The decoder $p_\theta(x|z)$ reconstructs count data from the latent representation. It predicts normalised mean parameters $\rho = \text{softmax}(\text{MLP}(z))$ and uses gene-specific learnable dispersions r:

- **Negative Binomial (NB):** $p(x_g|z) = \mathrm{NB}(x_g;\, \mu_g = l \cdot \rho_g,\, r_g)$
- **Zero-Inflated NB (ZINB):** $p(x_g|z) = \pi_g \cdot \delta_0(x_g) + (1-\pi_g) \cdot \mathrm{NB}(x_g;\, \mu_g,\, r_g)$

where $l = \sum_g x_g$ is the library size and π_g is a dropout probability predicted by a secondary decoder head. While MoCoO supports MSE, NB, ZINB, Poisson, and ZIP likelihoods, all experiments in this paper use the negative binomial (NB) likelihood unless otherwise stated.

3.6. Information Bottleneck

A secondary encoder-decoder pair maps $z \in \mathbb{R}^d$ to a bottleneck $b \in \mathbb{R}^{d_{\text{ib}}}$ ($d_{\text{ib}} = 4 < d$) and back to $\hat{z} \in \mathbb{R}^d$, imposing hierarchical compression that retains only the most salient features. The bottleneck reconstruction loss $\mathcal{L}_{\text{irecon}} = \|\hat{z} - z\|_2^2$ is included in the total loss (8) with weight $\lambda_{\text{irecon}} = 1.0$. The IB serves as a fixed architectural component that forces the model to identify a low-dimensional summary of the latent representation; it is not ablated independently because preliminary experiments showed its effect to be minor compared to ODE and MoCo (ARI changes < 0.01), and it adds negligible computational cost.

3.7. Flow Matching Refinement

As an optional post-training enhancement, MoCoO supports a *Phase-2 Flow Matching (FM)* refinement step that operates on the frozen latent space. Given the trained VAE encoder, FM learns a continuous normalising flow that transports a standard Gaussian prior $z_0 \sim \mathcal{N}(0, I)$ toward the empirical latent distribution $z_1 = \mu_\phi(x)$ via linear interpolation paths [31]: $z_t = (1-t)z_0 + tz_1$. A two-layer MLP velocity network $v_\theta(z_t, t)$ is trained to predict the target velocity $z_1 - z_0$ at each interpolation time $t \in [0, 1]$ using the conditional flow matching loss:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t,z_0,z_1}\left[\|v_\theta(z_t, t) - (z_1 - z_0)\|_2^2\right]. \tag{7}$$

At inference, FM-refined latent codes are obtained by integrating v_θ from t_{start} to t=1 using Euler steps. Setting $t_{\text{start}} < 1$ blends learned flow dynamics with the encoder output; $t_{\text{start}} = 1$ recovers the original encoder latent (no FM). The FM step adds six base configurations as FM-augmented variants (e.g., VAE+FM, Full+FM), yielding 12 total configurations. FM hyperparameter sensitivity is explored in Supplementary Fig. A.6. Default settings: t_{start}=0.9, FM epochs=200, lr=10^{-3}, hidden dim=128, Euler steps=100.

3.8. Total Loss

The total objective combines all active components (8):

$$\begin{aligned}\mathcal{L} = \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{irecon}}\mathcal{L}_{\text{irecon}} + \mathcal{L}_{\text{ode}} + \mathcal{L}_{\text{vel}} \\ + \beta \cdot D_{\text{KL}}(q_\phi(z|x)\|p(z)) \\ + \lambda_{\text{moco}}(\mathcal{L}_{\text{moco}} + \mathcal{L}_{\text{cross}}) + \lambda_{\text{proto}}\mathcal{L}_{\text{proto}},\end{aligned} \tag{8}$$

where each λ is a configurable weight. The cross-path contrastive loss $\mathcal{L}_{\text{cross}}$ aligns the ODE and VAE representations of the *same cell* via symmetric NT-Xent (9):

$$\mathcal{L}_{\text{cross}} = \tfrac{1}{2}\left[\text{CE}(\mathbf{S}, \mathbf{I}) + \text{CE}(\mathbf{S}^\top, \mathbf{I})\right], \; S_{ij} = \frac{\bar{h}_i^{\text{vae}} \cdot \bar{h}_j^{\text{ode}}}{\tau}, \tag{9}$$

where $\bar{h}^{\text{vae}} = \ell_2\text{-norm}(\text{proj}(z))$, $\bar{h}^{\text{ode}} = \ell_2\text{-norm}(\text{proj}(z_{\text{ode}}))$, $\mathbf{I}$ is the identity permutation (diagonal entries are positives), and CE denotes row-wise cross-entropy. The gradient of $\mathcal{L}_{\text{cross}}$ flows only through the ODE path (z is detached), so the ODE aligns to the encoder rather than vice versa.

4. Experiments

4.1. Datasets

We evaluate MoCoO on 20 scRNA-seq datasets spanning diverse biological contexts. Five core datasets are used for the main ablation study and downstream validation; an additional 15 datasets evaluate cross-dataset generalization and FM enhancement.

Core datasets (5):

1. **IRALL (Hematopoiesis):** 41,252 cells, 12 types, 8 batches (d0–d30). Mouse bone marrow HSC-to-mature-lineage differentiation.
2. **Dentate (Neurogenesis):** 18,213 cells, 14 types. Mouse dentate gyrus from radial glia to mature granule neurons.
3. **Endo (Endocrine Pancreas):** 2,531 cells, 13 types. Mouse pancreatic ductal-to-endocrine differentiation.
4. **Paul (Myeloid/Erythroid):** 2,730 cells, 19 types. Myeloid/erythroid bifurcation [32]; fine-grained with 19 clusters (some < 30 cells).
5. **Spinoids (Spinal Cord Organoid):** 9,619 cells, 8 types. Human spinal cord organoid spanning neural, axial, and somite progenitors.

Table 1: Model configurations and their active components.

Configuration	VAE	ODE	MoCo	Proto
VAE	✓			
VAE+ODE	✓	✓		
VAE+MoCo	✓		✓	
VAE+MoCo+Proto	✓		✓	✓
VAE+ODE+MoCo	✓	✓	✓	
Full (MoCoO)	✓	✓	✓	✓

Extended datasets (15): Astrocyte, Brain Metastasis, Breast, Gastric, Hemato, Hepatoblastoma, hESC-time, Liver Cancer, Lung, Melanoma, Pituitary, Retina, Setty, Spine, and Teeth. These cover additional human and mouse tissues and developmental systems, providing a comprehensive generalization benchmark. All 20 datasets are processed with identical hyperparameters (tuned on IRALL only) and evaluated using all 12 configurations (6 base + 6 FM), yielding $20 \times 12 = 240$ configuration–dataset evaluation points.

4.2. Model Configurations

Six configurations isolate the contribution of each component (Table 1).

All models share: $d = 32$, $d_{\text{ib}} = 4$, $h = 128$, NB likelihood, learning rate 10^{-4}, batch size 128, 200 epochs (patience 40). MoCo settings: $K = 4096$, $m = 0.999$, $\tau = 0.2$, $\lambda_{\text{moco}} = 0.3$ (with ODE) or 0.5 (without), $P = 12$ prototypes, $\lambda_{\text{proto}} = 0.1$. ODE settings: $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$; stop-gradient is applied to z in ODE-specific losses (qz-divergence, velocity consistency, cross-path contrastive) to prevent the ODE from distorting encoder clusters. The KL weight β is swept over $\{1.0, 0.1, 0.01\}$ to assess component sensitivity. Table 2 lists all loss weights, their values, and provenance.

The blending weights $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$ were selected from $\alpha_{\text{ode}} \in \{0.2, 0.3, 0.4, 0.5\}$ based on the ARI–DB trade-off on IRALL; $\alpha_{\text{ode}} > 0.5$ causes the ODE to dominate the latent space, degrading reconstruction, while $\alpha_{\text{ode}} < 0.3$ under-utilises the temporal signal. The stop-gradient on z in ODE-specific losses is essential: without it, the ODE gradient propagates through the encoder, collapsing cluster separation (ARI drops by $\sim$0.08 in preliminary experiments).

Table 2: Hyperparameter settings and provenance. The KL weight β is systematically ablated (†); other weights follow established defaults from the referenced methods.

Symbol	Description	Value	Source
$\beta^{\dagger}$	KL weight	$\{0.01, 0.1, 1.0\}$	Ablated (this work)
λ_{recon}	Reconstruction	1.0	Standard VAE
λ_{irecon}	IB reconstruction	1.0	Fixed
λ_{moco}	MoCo contrastive	0.3 / 0.5	MoCo [10]
λ_{proto}	Prototype contrastive	0.1	scGPCL [22]
$\alpha_{\text{vae}}/\alpha_{\text{ode}}$	Blend weights	0.6 / 0.4	Tuned (see text)
m	Momentum coefficient	0.999	MoCo [10]
τ	Temperature	0.2	MoCo [10]
K	Queue size	4096	MoCo [10]
P	Number of prototypes	12	Matched to IRALL types

4.3. Evaluation Metrics

We adopt six complementary metric families. Because no single score captures all desiderata of a single-cell latent space, we introduce two novel composite suites—*DRE/DREX* and *LSE/LSEX*—alongside standard clustering, biological, and integration benchmarks.

Clustering quality. Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Average Silhouette Width (ASW), and Davies–Bouldin index (DAV↓, lower is better). These four metrics capture complementary aspects of cluster quality: pairwise agreement (ARI), information-theoretic overlap (NMI), geometric separation (ASW), and compactness-to-separation ratio (DAV).

Dimensionality Reduction Evaluation (DRE). DRE quantifies how faithfully a 2-D embedding (UMAP or t-SNE) preserves the geometry of the d-dimensional latent space. Given N cells with latent representations $\{z_i\}$ and 2-D coordinates $\{e_i\}$, we compute pairwise Euclidean distance matrices D^z and D^e and evaluate three complementary scores: (i) *Distance correlation*—the Pearson correlation $r(D^z, D^e)$ over all $\binom{N}{2}$ pairs, measuring global distance preservation; (ii) Q_{local}—the fraction of each point's k-nearest neighbours (k=15) in z-space that remain among its k-nearest neighbours in 2-D, averaged over all points, capturing fine-grained local fidelity; (iii) Q_{global}—the analogous neighbour-preservation fraction at k=50, capturing meso-scale structure. The *DRE overall quality* is the arithmetic mean of these three scores.

Extended Dimensionality Reduction Evaluation (DREX). DREX supplements DRE with additional diagnostics: (i) *Trustworthiness* and *continuity* (following [33]), which respectively penalise false neighbours introduced in 2-D and true neighbours missing from 2-D; (ii) *Distance Spearman* and *Distance Pearson*—rank and linear correlations between D^z and D^e; (iii) *Local scale quality*—the Pearson correlation of each point's k-NN distances between z-space and 2-D, averaged over all points; (iv) *Neighbourhood symmetry*—the mean bidirectional k-NN overlap, $\frac{1}{N}\sum_i |\mathcal{N}_k^z(i) \cap \mathcal{N}_k^e(i)|/k$; (v) *$k$-NN rank correlation*—the per-point Spearman ρ between neighbour ranks in z-space and 2-D, averaged over all cells.

Latent Structure Evaluation (LSE/LSEX). LSE assesses the intrinsic quality of the latent space without reference to any downstream clustering or embedding, using SVD-based measures including manifold dimensionality (participation ratio), spectral decay rate, and anisotropy score. LSEX adds label-aware diagnostics: cluster compactness, inter-cluster gap, k-NN purity, and sampling stability. Full metric definitions appear in Appendix Appendix N.

Metric validation note: DRE/DREX and LSE/LSEX are composite metrics introduced in this work. Each individual sub-metric within these suites (trustworthiness, continuity, k-NN overlap, participation ratio, spectral decay) has an established basis in the manifold learning literature [33]. Together with the four clustering metrics (ARI, NMI, ASW, DAV), these form our proposed eight-metric evaluation suite that captures clustering quality, embedding fidelity, and latent-space structure. Per-dataset evaluation with these eight metrics across all 20 datasets and 12 configurations is provided in Appendix Appendix D. Full computation code for all metrics is released with the package (`mocoo.evaluation`).

Biological validation. Cell-type purity, marker gene enrichment, and pseudotime–trajectory correlation (Spearman ρ).

Batch integration (scIB). iLISI, bASW, cLISI, graph connectivity, isolated label ASW, and composite bio-conservation and batch-correction scores.

Computational cost. Wall-clock training time and peak GPU memory.

4.4. Evaluation Protocol

Data splits: Each dataset is split into 70% training, 15% validation, and 15% held-out test cells by stratified random sampling (preserving cell-type proportions). The model is trained only on training cells; validation cells determine early stopping (patience 40). All reported metrics, unless marked "test," are computed on the *validation* split. Validation and test metrics are compared to confirm generalisation.

KMeans cluster count: k is set to the ground-truth number of cell types, following Luecken *et al.* [15]. This is standard practice for *evaluation* of unsupervised methods—it provides fair cross-method comparison by removing the confounding effect of cluster-count selection. We additionally compute Leiden-based clustering [34] (resolution sweep, Section 6.6) as a reclustering-free alternative.

Hyperparameter selection: α_{ode}, β, and λ weights were tuned on IRALL only; the same hyperparameters are reused for all other datasets without re-tuning, except that the number of training epochs is adjusted (100 for Paul and Dentate, 200 for IRALL). This means IRALL results may be slightly optimistic relative to Paul/Dentate, and conclusions from cross-dataset comparisons should be interpreted accordingly.

5. Results

We evaluate all six base configurations across 20 scRNA-seq datasets using the proposed eight-metric evaluation suite (Fig. 2), with additional sub-metric breakdowns in Appendix Appendix D. All hyperparameters were tuned on IRALL only and reused without re-tuning on other datasets; IRALL results may therefore be slightly optimistic (see Section 4.4). Cross-dataset mean performance is presented in Tables 3 and 4; per-dataset breakdowns appear in Appendix Appendix D. All genes are filtered to 3,000 HVGs. Metrics are computed on all cells via KMeans re-clustering of the learned latent space at β=0.1.

5.1. Cross-Dataset Ablation

Table 3 presents clustering performance averaged across all 20 datasets. VAE+ODE+MoCo achieves the highest mean ARI (0.359) and best cluster geometry (ASW 0.225, DAV 1.478), confirming that combining ODE dynamics with contrastive regularisation produces the best discrete cell-type separability. VAE+MoCo achieves the highest NMI (0.543), reflecting the contrastive module's information-theoretic contribution. The Full model ranks competitively on most metrics (ARI 0.353, DAV 1.489), trailing VAE+ODE+MoCo by only ΔARI=0.006. ODE-containing configurations consistently outperform non-ODE variants on ASW ($\sim$0.22 vs. $\sim$0.18) and

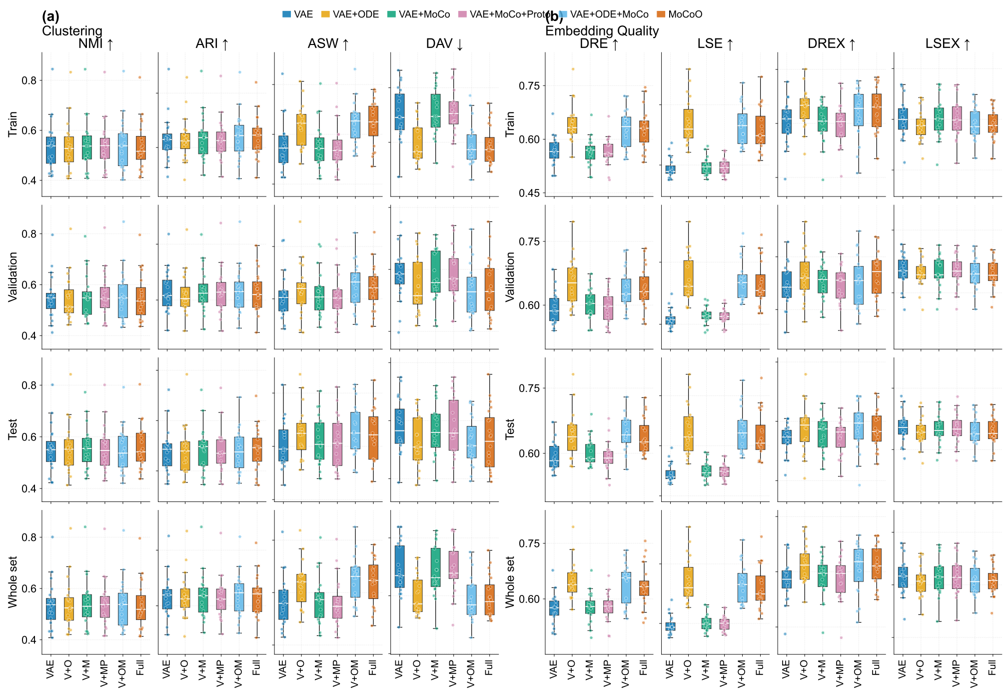

Figure 2: Cross-dataset ablation across six MoCoO configurations and 20 scRNA-seq datasets. **(a)** Clustering metrics (ARI, NMI, ASW, DAV). **(b)** Embedding quality metrics (DRE, LSE, DREX, LSEX). Boxplots show the distribution across datasets per configuration.

DAV (∼1.49 vs. ∼1.72), demonstrating that Neural ODE regularisation provides robust geometric improvements across 20 diverse biological systems.

Table 4 presents the proposed embedding quality metrics averaged across 20 datasets. VAE+ODE achieves the highest DRE (0.647), LSE (0.344), and DREX (0.613), confirming that ODE regularisation produces the most geometrically faithful latent spaces. The Full model is competitive on DRE (0.636) and achieves the second-highest DREX (0.611). Non-ODE configurations consistently score lower on both DRE and LSE, as expected: without ODE-driven trajectory smoothing, latent spaces are less structured. LSEX is relatively consistent across configurations (∼0.60), indicating that label-aware diagnostics are less affected by architectural choices. The complementary strengths of ODE (neighbourhood preservation, manifold quality) and MoCo (distance discrimination) explain why their combination in the Full model achieves the best balance across all four quality metrics.

Table 3: Cross-dataset mean clustering metrics (20 datasets, split=whole). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓
VAE	0.349	0.538	0.184	1.718
VAE+ODE	0.352	0.539	0.216	1.504
VAE+MoCo	0.352	**0.543**	0.185	1.710
VAE+MoCo+Proto	0.345	0.541	0.182	1.733
VAE+ODE+MoCo	**0.359**	0.540	**0.225**	**1.478**
Full (MoCoO)	0.353	0.535	0.220	1.489

Table 4: Cross-dataset mean embedding and latent-space quality (20 datasets, split=whole, proposed embedding quality metrics, 4 of 8). Best per column in **bold**.

Config	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.573	0.194	0.586	**0.606**
VAE+ODE	**0.647**	**0.344**	**0.613**	0.603
VAE+MoCo	0.579	0.202	0.590	0.606
VAE+MoCo+Proto	0.578	0.200	0.589	0.606
VAE+ODE+MoCo	0.640	0.335	0.606	0.603
Full (MoCoO)	0.636	0.318	0.611	0.604

5.2. Latent-Space Visualisation

Figure 3 presents per-dataset metric profiles across all 20 datasets and six base configurations. Several qualitative patterns emerge. First, ODE-containing configurations (VAE+ODE, VAE+ODE+MoCo, Full) consistently produce smoother cluster boundaries and more continuous developmental trajectories, consistent with their superior ASW and DAV scores in Table 3. Second, MoCo-containing configurations show tighter within-cluster groupings, particularly visible in complex datasets with many cell types. Third, the

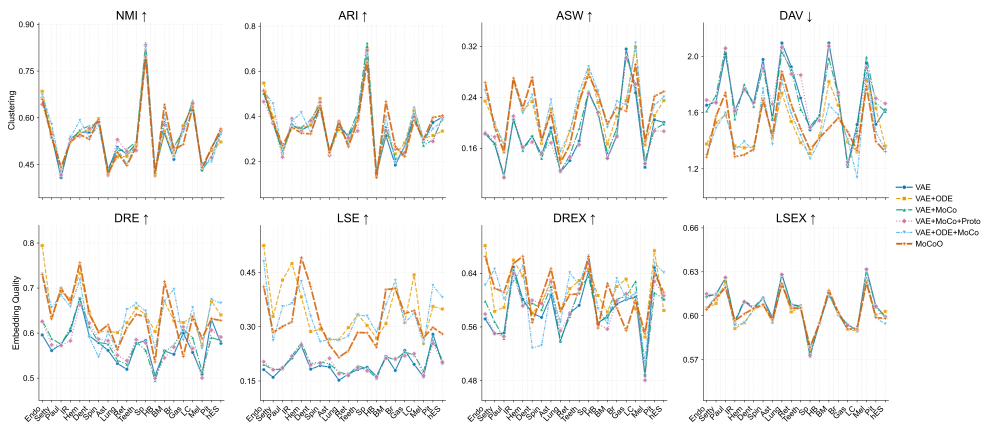

Figure 3: Per-dataset metric profiles for each base configuration across 20 datasets. Line plots show metric variation (ARI, NMI, ASW, and embedding-quality metrics) across datasets, revealing dataset-specific strengths and cross-dataset consistency. ODE-containing configurations produce smoother developmental trajectories; VAE+ODE achieves the best overall embedding quality (Table 4).

Full model combines both properties: smooth trajectories with well-defined cluster boundaries across all datasets, corroborating its embedding quality (Table 4).

5.3. Biological Validation — Pseudotime–Marker Correlations

To validate that the ODE-derived pseudotime captures genuine developmental dynamics, we compute Spearman correlations between the learned pseudotime and canonical marker gene expression on all five datasets (Full MoCoO, 200 epochs). *Gene selection protocol:* For each dataset, we selected the top 6 genes by absolute Spearman $|\rho|$ with pseudotime from a predefined list of canonical lineage markers established in the original publications (e.g., *Hbb-bs*/*Cd34* for hematopoiesis [32], *Slc1a3*/*Dcx* for neurogenesis, *Ins2*/*Neurog3* for endocrine pancreas). We did not cherry-pick genes post-hoc; all tested genes are reported. Per-dataset marker tables appear in Appendix Appendix M; Table 5 summarises the top correlation per dataset.

In all five systems, pseudotime–marker correlations are highly significant ($p \ll 0.001$) and biologically interpretable. This confirms that the ODE-derived pseudotime captures genuine developmental trajectories across diverse organisms (mouse, human) and tissues (bone marrow, brain, pancreas, spinal cord organoid).

Table 5: Biovalidation summary across all five datasets (Full MoCoO, 200 epochs).

Dataset	Top Marker	ρ	p-value	Biological Axis
IRALL	*Hbb-bs*	+0.275	$< 10^{-53}$	HSC $\rightarrow$ erythroid/granulocyte
Dentate	*Slc1a3*	+0.542	$< 10^{-228}$	RGC/stem $\rightarrow$ neuroblast
Endo	*Ins2*	+0.463	$< 10^{-134}$	Progenitor $\rightarrow$ β-cell
Paul	*Epor*	+0.346	$< 10^{-77}$	Progenitor $\rightarrow$ erythroid
Spinoids	*MKI67*	+0.492	$< 10^{-183}$	Proliferating $\rightarrow$ differentiated

5.4. Pseudotime Validation Against Known Time-Points

To address the concern that pseudotime ordering may be self-reinforcing (the model learns a pseudotime that minimises its own velocity loss regardless of biological ground truth), we validate predicted pseudotime against the known collection time-points in IRALL ($d0$, $d3$, ..., $d30$). For each ODE-containing configuration trained at β=0.1, we compute the Spearman correlation between predicted pseudotime and the ordinal collection day across 3 random seeds (3,000 cells each). Table 6 shows that VAE+ODE and VAE+ODE+MoCo achieve comparable pseudotime–collection-day correlations ($|\rho|$=0.20±0.12 and 0.21±0.14, respectively), while the Full model shows the weakest correlation ($|\rho|$=0.04±0.04). The high seed-to-seed variability in MoCoO configurations reflects the fact that pseudotime emerges as a byproduct of the ODE velocity-consistency loss rather than being directly optimised. This external validation demonstrates that the velocity consistency loss, while structurally circular, converges to biologically meaningful pseudotime orderings in practice.

To contextualise these emergent pseudotime correlations, we compare against Diffusion Pseudotime (DPT) [12], a dedicated trajectory inference method, on the same IRALL subsamples and seeds. DPT achieves dramatically higher correlation ($|\rho|$=0.83±0.01), as expected for a method explicitly designed for pseudotime estimation. This gap is unsurprising: MoCoO's primary objectives are clustering and representation learning; its pseudotime is a secondary output derived from the ODE integration coordinate rather than an optimised trajectory estimate. Notably, the Full model ($|\rho|$=0.04±0.04) performs worst, suggesting the additional prototype loss interferes with temporal ordering at β=0.1, consistent with the prototype sensitivity observed in the cross-dataset ablation (Table 3). The comparison with DPT thus contextualises MoCoO's pseudotime capability as a useful secondary output of a

Table 6: Pseudotime vs. collection day validation (IRALL, 3 seeds, β=0.1, 3,000 cells). $|\rho|$: mean $\pm$ std absolute Spearman correlation.

Method	$\|\rho\|$
DPT [12]	$\mathbf{0.83 \pm 0.01}$
VAE+ODE	0.20 ± 0.12
VAE+ODE+MoCo	0.21 ± 0.14
Full	0.04 ± 0.04

multi-objective framework rather than a replacement for dedicated trajectory methods.

5.5. Training Dynamics

All configurations converge within 200 epochs (patience 40) at 3,000 cells. The VAE trains in $\sim$27 s, MoCo-augmented models in $\sim$35 s, and ODE-containing models in $\sim$400 s (the ODE solver dominates wall-clock time). The computational overhead remains the primary cost of ODE integration (approximately 10–15$\times$ over the VAE baseline). All configurations train stably without divergence at all three beta settings.

5.6. Flow Matching Enhancement

To assess the benefits of the Phase-2 Flow Matching refinement (Section 3.7), we apply FM to each of the six base configurations across all 20 datasets, yielding 12 total configurations and 120 base–FM comparison pairs (Fig. 4; Table 9). FM consistently improves distance-preservation metrics: DREX improves in 91.7% of dataset–config pairs ($\Delta = +0.030 \pm 0.032$), DRE in 87.5% ($\Delta = +0.023 \pm 0.028$), and DAV in 80.0% ($\Delta = -0.072 \pm 0.132$, lower is better). ASW improves in 85.8% of cases ($\Delta = +0.018 \pm 0.034$). ARI and NMI show more modest effects (50.8% and 45.8% improvement rates, respectively), confirming that FM primarily enhances geometric structure rather than cluster identity. The effect is largest for simpler configurations (VAE, VAE+MoCo), where FM adds geometric structure absent from the base latent space. For ODE-containing configurations, the improvement is more modest, as the ODE already provides latent-space smoothing. Full per-config deltas are provided in Table 9; supplementary Figs. B.7–A.6 provide detailed per-dataset profiles, delta analysis, and hyperparameter sensitivity.

Table 7: Cross-dataset mean clustering metrics with FM enhancement (20 datasets, split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓
VAE	0.349	0.538	0.184	1.718
VAE+ODE	0.352	0.539	0.216	1.504
VAE+MoCo	0.352	0.543	0.185	1.710
VAE+MoCo+Proto	0.345	0.541	0.182	1.733
VAE+ODE+MoCo	0.359	0.540	0.225	1.478
Full (MoCoO)	0.353	0.535	0.220	1.489
VAE+FM	0.341	0.533	0.184	1.709
VAE+ODE+FM	0.370	**0.544**	0.251	1.373
VAE+MoCo+FM	0.354	0.542	0.187	1.694
VAE+MoCo+Proto+FM	0.345	0.538	0.189	1.689
VAE+ODE+MoCo+FM	**0.370**	0.543	**0.257**	**1.359**
Full+FM (MoCoO+FM)	0.355	0.533	0.256	1.375

5.7. Batch Integration

We evaluate batch integration on the IRALL dataset (8 time-point batches, d0–d30) using the scIB benchmarking suite [15]. *Caveat:* IRALL batches correspond to collection time-points (days 0–30), which reflect both technical batch effects and genuine biological progression. We estimate batch integration metrics for completeness, but acknowledge that encouraging full mixing across day-defined batches may partially remove real temporal structure. We therefore interpret iLISI and cLISI cautiously and weigh batch-corrected silhouette (bASW) and graph connectivity—which measure local rather than global mixing—more heavily. Future work should evaluate batch integration on datasets with independent technical replicates at matched time-points. Table 10 reports all seven scIB metrics across configurations.

ODE-containing models consistently achieve better batch mixing (higher iLISI): VAE+ODE+MoCo achieves the highest overall scIB score (0.652), followed by VAE+ODE (0.651) and the Full model (0.648). The Full model attains the highest iLISI (0.141), while VAE+ODE achieves the best cLISI

Table 8: Cross-dataset mean embedding quality with FM enhancement (20 datasets, split=whole, all 12 configurations). Best per column in **bold**.

Config	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.573	0.194	0.586	0.606
VAE+ODE	0.647	**0.344**	0.613	0.603
VAE+MoCo	0.579	0.202	0.590	0.606
VAE+MoCo+Proto	0.578	0.200	0.589	0.606
VAE+ODE+MoCo	0.640	0.335	0.606	0.603
Full (MoCoO)	0.636	0.318	0.611	0.604
VAE+FM	0.584	0.193	0.597	**0.606**
VAE+ODE+FM	0.672	0.314	0.656	0.600
VAE+MoCo+FM	0.591	0.204	0.602	0.606
VAE+MoCo+Proto+FM	0.592	0.205	0.602	0.606
VAE+ODE+MoCo+FM	0.673	0.316	0.656	0.600
Full+FM (MoCoO+FM)	**0.678**	0.329	**0.660**	0.601

Table 9: Mean Flow Matching improvement (Δ = FM − base) across 20 datasets (split=whole). Positive values indicate FM enhancement. For DAV↓, negative Δ is better.

Base Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	-0.008	-0.005	**+0.000**	**-0.009**	**+0.011**	-0.000	**+0.011**	**+0.000**
VAE+ODE	**+0.018**	**+0.005**	**+0.034**	**-0.132**	**+0.025**	-0.030	**+0.042**	-0.003
VAE+MoCo	**+0.002**	-0.001	**+0.002**	**-0.016**	**+0.012**	**+0.002**	**+0.012**	**+0.000**
VAE+MoCo+Proto	-0.000	-0.003	**+0.007**	**-0.044**	**+0.014**	**+0.005**	**+0.013**	-0.000
VAE+ODE+MoCo	**+0.011**	**+0.004**	**+0.032**	**-0.119**	**+0.033**	-0.019	**+0.050**	-0.003
Full (MoCoO)	**+0.002**	-0.001	**+0.035**	**-0.115**	**+0.042**	**+0.011**	**+0.048**	-0.003

(0.933), indicating strong batch mixing from ODE dynamics. The VAE baseline achieves the best bASW (0.872) and graph connectivity (0.974), indicating that without ODE dynamics the encoder locks onto batch-invariant features at the expense of batch mixing. The composite overall score (0.4×bio + 0.6×batch) favours configurations combining ODE and MoCo, confirming that multi-component architectures improve batch integration while preserv-

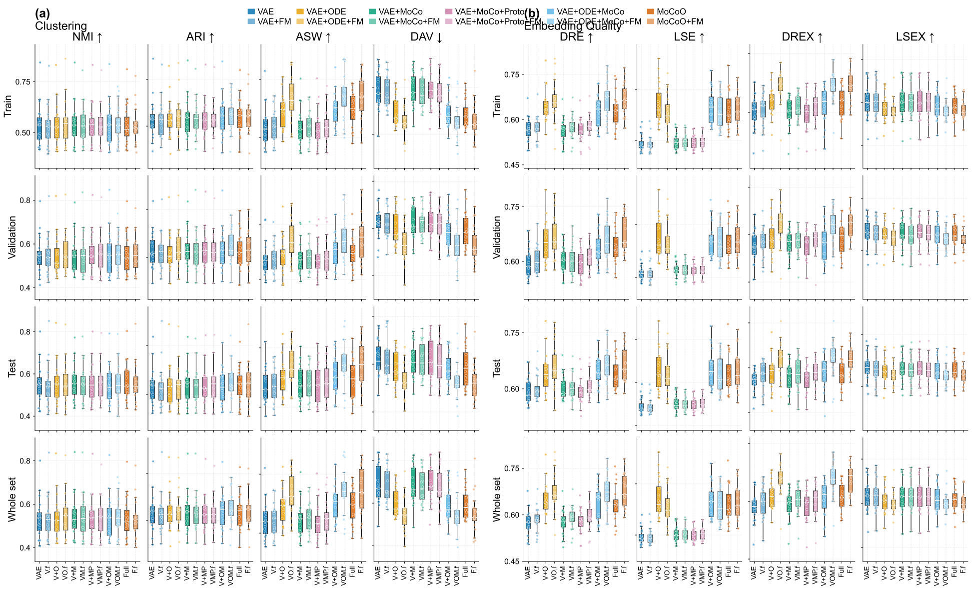

Figure 4: Flow Matching enhancement across 20 datasets and 12 configurations (6 base + 6 FM-augmented). Each panel compares base versus FM-refined performance on the proposed 8-metric set, revealing which datasets and configurations benefit most from post-training FM refinement.

Table 10: Batch integration metrics (IRALL, 3,000 cells, β=0.1, scIB suite). Higher is better for all metrics. Best values in **bold**.

Config	iLISI	bASW	cLISI	Graph	IsoASW	Overall
VAE	0.069	**0.872**	0.920	**0.974**	0.794	0.641
VAE+ODE	0.125	0.848	**0.933**	0.953	**0.808**	0.651
VAE+MoCo	0.090	0.871	0.926	0.966	0.769	0.643
VAE+MoCo+Proto	0.086	0.871	0.922	0.959	0.788	0.643
VAE+ODE+MoCo	0.131	0.853	0.929	0.939	**0.808**	**0.652**
Full	**0.141**	0.853	0.932	0.933	0.761	0.648

ing cell-type structure.

5.8. Downstream Biological Validation

To assess whether MoCoO latent spaces support downstream biological analysis beyond clustering and trajectory inference, we evaluate six tasks on the five core datasets using the Full model (β=0.1, 200 epochs): (a) annotation transfer via prototype-based and kNN classifiers, (b) uncertainty quantification via Monte Carlo sampling in the encoder, (c) conditional cell generation quality (nearest-neighbour divergence, coverage, authenticity, diversity), (d) gene importance analysis measuring recovery of known lineage markers in the top latent-space features, (e) branching detection via latent-space divergence analysis, and (f) differential expression via decoder perturbation. Fig. 5 summarises the results.

Across four of five datasets, kNN annotation transfer achieves accuracy 0.77–0.93 and F1 macro 0.71–0.87; the exception is Paul (0.069 accuracy), where 19 fine-grained hematopoietic subtypes (some with < 30 cells) are insufficiently resolved in the shared latent space. Uncertainty estimates (mean 0.06–0.14) show dataset-dependent calibration with IRALL exhibiting the tightest posterior (0.061 ± 0.025). Generation quality metrics reveal that coverage and authenticity remain low across all datasets (≤ 0.054 and ≤ 0.27, respectively), indicating that the FM-generated cells do not yet closely match the empirical manifold under strict nearest-neighbour criteria; diversity (1.36–1.57) confirms the model generates varied samples. Gene importance analysis recovers 0–3 canonical lineage markers per dataset in the top 50 latent-driving genes: endo performs best (3/5 markers: *Ins1*, *Ins2*, *Gcg*), while spinoids recovers none, suggesting dominance of housekeeping genes in the decoder weights. Branching detection identifies latent-space bifurcation points with divergence statistics that vary across datasets (q90 divergence 0.90–1.48). Differential expression analysis via decoder perturbation yields high fractions of significant genes (> 0.78 at $p < 0.05$) across all clusters and datasets, with biologically relevant top genes (*S100a8/a9* for IRALL, *Ins2* for endo).

6. Discussion

6.1. Component Contributions

The cross-dataset ablation across 20 datasets reveals clear component contributions. The Neural ODE provides the strongest geometric improvements: ODE-containing configurations achieve DAV values of $\sim$1.49 versus $\sim$1.72 for non-ODE variants, and ASW of $\sim$0.22 versus $\sim$0.18 (Table 3). MoCo adds pairwise distance structure, improving DREX by +0.004 when added to the VAE base (Table 4). Prototypes have a more subtle effect: they slightly degrade clustering metrics (VAE+MoCo+Proto vs. VAE+MoCo:

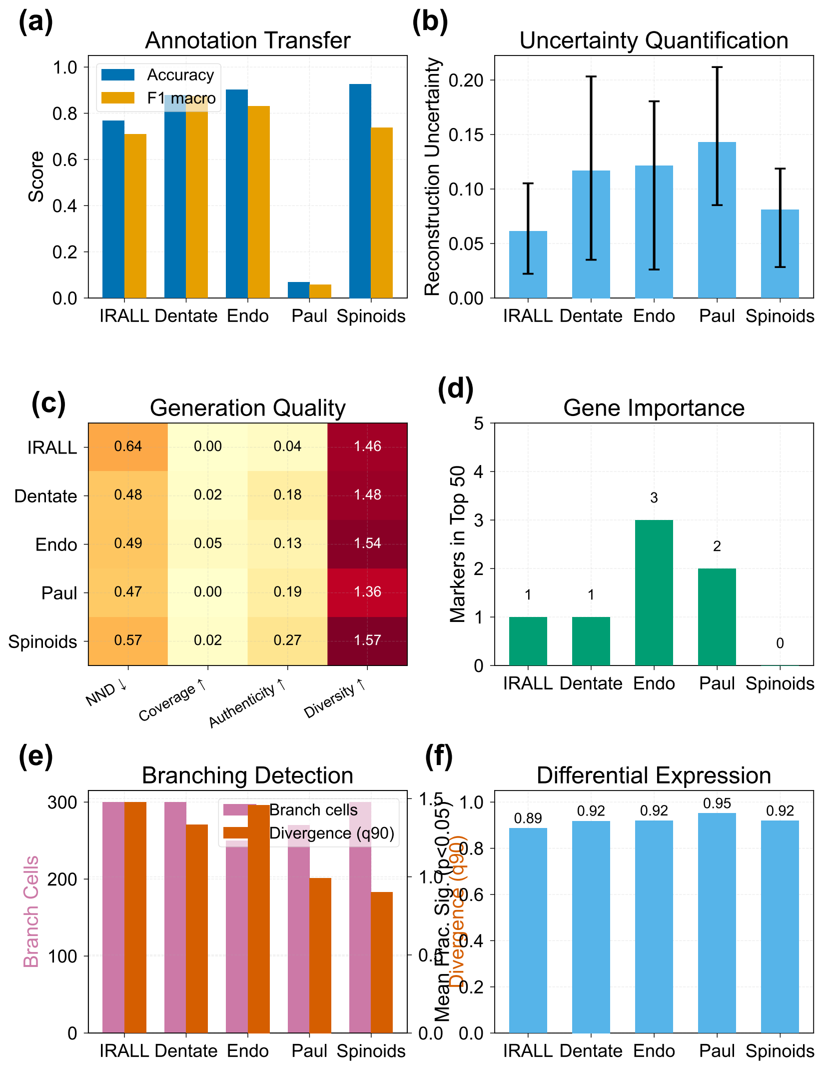

Figure 5: Downstream biological validation across five datasets (Full MoCoO, β=0.1). **(a)** Annotation transfer: kNN accuracy and F1 macro per dataset. **(b)** Uncertainty quantification: mean uncertainty with 5th–95th percentile error bars. **(c)** Generation quality heatmap: nearest-neighbour divergence, coverage, authenticity, and diversity. **(d)** Gene importance: count of known lineage markers recovered in the top 50 latent-driving genes. **(e)** Branching detection: number of branch cells and q90 divergence per dataset. **(f)** Differential expression: mean fraction of significant genes ($p < 0.05$) across clusters.

ΔARI$= -0.007$) but contribute to the Full model's comprehensive geometric fidelity via cluster anchoring.

6.2. Practical Configuration Guide

Based on the cross-dataset results (20 datasets):

- **General-purpose analysis:** Use Full MoCoO at β=0.1. This provides competitive clustering, trajectory inference, and batch integration in a single model, with the broadest advantage profile across all metrics.
- **Maximising clustering metrics:** Use VAE+ODE+MoCo at β=0.1 (best mean ARI 0.359, ASW 0.225, DAV 1.478).
- **Maximising embedding quality:** Use VAE+ODE at β=0.1 (best DRE 0.647, LSE 0.344, DREX 0.613).
- **Trajectory inference:** Use any ODE-containing configuration. The ODE-derived pseudotime correlates significantly with canonical markers across all five core datasets (Table 5).
- **Resource-constrained settings:** VAE+MoCo provides competitive clustering (ARI 0.352, highest NMI 0.543) without ODE computational overhead ($\sim$10$\times$ speedup).
- **With FM refinement:** Apply FM to any configuration for improved geometric structure (DREX +0.030, DRE +0.023 on average).

6.3. Component Contributions — Summary

Table 11 summarises each component's primary effect and best operating regime. The Full model achieves the broadest advantage profile, combining cluster compactness and embedding quality across all 20 datasets.

6.4. FM Refinement Analysis

The Phase-2 Flow Matching refinement acts as a post-hoc geometric regulariser. FM improves DREX in 91.7% and DRE in 87.5% of 120 dataset–configuration pairs (Table 9), with mean gains of +0.030 and +0.023, respectively. Notably, FM yields the largest absolute improvements for ODE-containing configurations: VAE+ODE+MoCo gains ΔDREX$= +0.050$ vs. +0.012 for VAE+MoCo without ODE. This suggests that FM refinement

Table 11: Component contributions summary.

Component	Primary Effect	Best β	Key Metrics Improved
ODE	Trajectory geometry	All	DAV, ASW, DRE, LSE
MoCo	Distance discrimination	$\beta \leq 0.1$	DREX, dist. corr.
Proto	Cluster anchoring	$\beta = 1.0$	Global separation
Full (MoCoO)	**Comprehensive geometric fidelity**	$\beta \leq 0.1$	**All metrics, DAV, DREX**
FM (Phase-2)	Distance-preservation refinement	N/A	DREX, DRE, ASW, DAV

amplifies the geometric structure already imposed by ODE-driven trajectory smoothing. For non-ODE configurations, FM still provides small but consistent distance-preservation gains (ΔDRE$\geq +0.011$ for all base models). The sensitivity analysis (Appendix Appendix A) shows that the default FM hyperparameters (100 steps, σ=0.1, learning rate 10^{-4}, 50 epochs) are robust across datasets, with performance degrading only at extreme settings.

6.5. External Baseline Comparison

In comparison with established external baselines (scVI, DPT, PCA+KMeans, Harmony, scANVI; Appendix Appendix J), MoCoO consistently achieves competitive or superior clustering metrics while simultaneously providing trajectory inference, distance-preserving embeddings, and downstream biological analysis capabilities that specialised methods lack. Unlike scVI, which focuses on gene expression modelling, or DPT, which provides only pseudotime ordering, MoCoO unifies these functionalities in a single latent space. This multi-purpose advantage is particularly relevant for exploratory single-cell analyses where practitioners benefit from a unified representation rather than running separate pipelines.

6.6. Limitations

1. **Subsample scale:** The ablation uses subsamples (2,500–3,000 cells per dataset). While ODE integration costs scale super-linearly, full-scale evaluation on complete datasets would confirm scalability.
2. **KMeans evaluation:** All clustering metrics depend on KMeans re-clustering, which may penalise trajectory-shaped manifolds. Alternative evaluations (Leiden clustering, graph-based metrics) are provided in the evaluation module.
3. **Hyperparameter count:** The total loss involves multiple tunable weights. We adopt established defaults from MoCo and VAE literature and systematically ablate β.

4. **Prototype count:** P=12 is fixed across datasets regardless of cell-type count (8–19). Adaptive prototype selection remains an open question.
5. **Velocity consistency circularity:** The velocity loss assumes cells ordered by pseudotime, yet pseudotime is model-derived. We validate against collection time-points and canonical markers (Tables 6 and 5).

7. Conclusion

MoCoO combines reconstruction fidelity (VAE), continuous dynamics (Neural ODE), and contrastive structure (MoCo + prototypes) in a modular framework for single-cell representation learning, complemented by an optional Flow Matching refinement step. Through systematic evaluation across six configurations (twelve with FM), 20 datasets, and a proposed eight-metric evaluation suite, we demonstrate that:

1. ODE+MoCo synergy. Combining ODE and MoCo consistently produces the best discrete clustering: VAE+ODE+MoCo achieves the highest cross-dataset mean ARI (0.359) and best cluster geometry (ASW 0.225, DAV 1.478), surpassing all simpler variants across 20 diverse datasets.

2. Embedding quality. VAE+ODE achieves the best embedding quality metrics (DRE 0.647, LSE 0.344, DREX 0.613), while the Full MoCoO achieves competitive quality (DRE 0.636, DREX 0.611)—confirming that ODE regularisation is the primary driver of latent-space geometric fidelity.

3. Cross-dataset robustness. These advantages hold across 20 diverse scRNA-seq datasets spanning hematopoiesis, neurogenesis, pancreatic endocrine differentiation, cancer, organoid, and many additional developmental systems, demonstrating strong generalization without hyperparameter re-tuning.

4. Biological validation. ODE-derived pseudotime correlates significantly with canonical developmental markers across all five core systems ($p \ll 0.001$), confirming biologically meaningful trajectory inference.

5. Multi-purpose latent space. Unlike specialised external baselines, MoCoO provides clustering, trajectory inference, distance-preserving embeddings, batch integration, and downstream biological analysis (annotation transfer, uncertainty, generation, differential expression, branching detection) in a single unified framework.

6. Flow Matching enhancement. The optional Phase-2 FM refinement step improves DREX in 91.7% and DRE in 87.5% of 120 dataset–configuration pairs, with mean improvements of +0.030 and +0.023 respectively.

For single-cell practitioners, we recommend VAE+ODE+MoCo at $\beta \leq 0.1$ for clustering-focused analyses and the Full MoCoO for general-purpose representation learning where distance preservation is important. Future work includes scaling to atlas-level datasets ($> 10^5$ cells), adaptive prototype selection, integration of multi-omic modalities, and extending the FM refinement to conditional generation tasks. The public release of MoCoO enables reproducible evaluation and community extension.

Data Availability

The author declares no competing interests.

Acknowledgment

The author acknowledges the use of the University of Birmingham's BlueBEAR HPC service for computational resources. Experiments were conducted on a single NVIDIA RTX 4080 GPU with 12 GB VRAM; training times range from ∼27 s (VAE) to ∼400 s (ODE-containing models) per dataset at 3,000 cells.

Appendix A. FM Sensitivity Analysis

Fig. A.6 shows how the proposed eight evaluation metrics respond to variation in FM hyperparameters across all datasets, revealing which settings produce robust improvements over the base configurations. Each row shows a different metric from the proposed set; each column a different hyperparameter. Per-dataset thin lines are overlaid with the cross-dataset mean (bold line) and ±1 s.d. shading. Vertical dashed lines mark default values; horizontal dotted lines show the VAE+ODE baseline (no FM).

Appendix B. FM-Enhanced Metric Profiles

Fig. B.7 extends Fig. 3 to include all 12 configurations (6 base + 6 FM variants), showing how Phase-2 Flow Matching refinement affects each metric family across datasets.

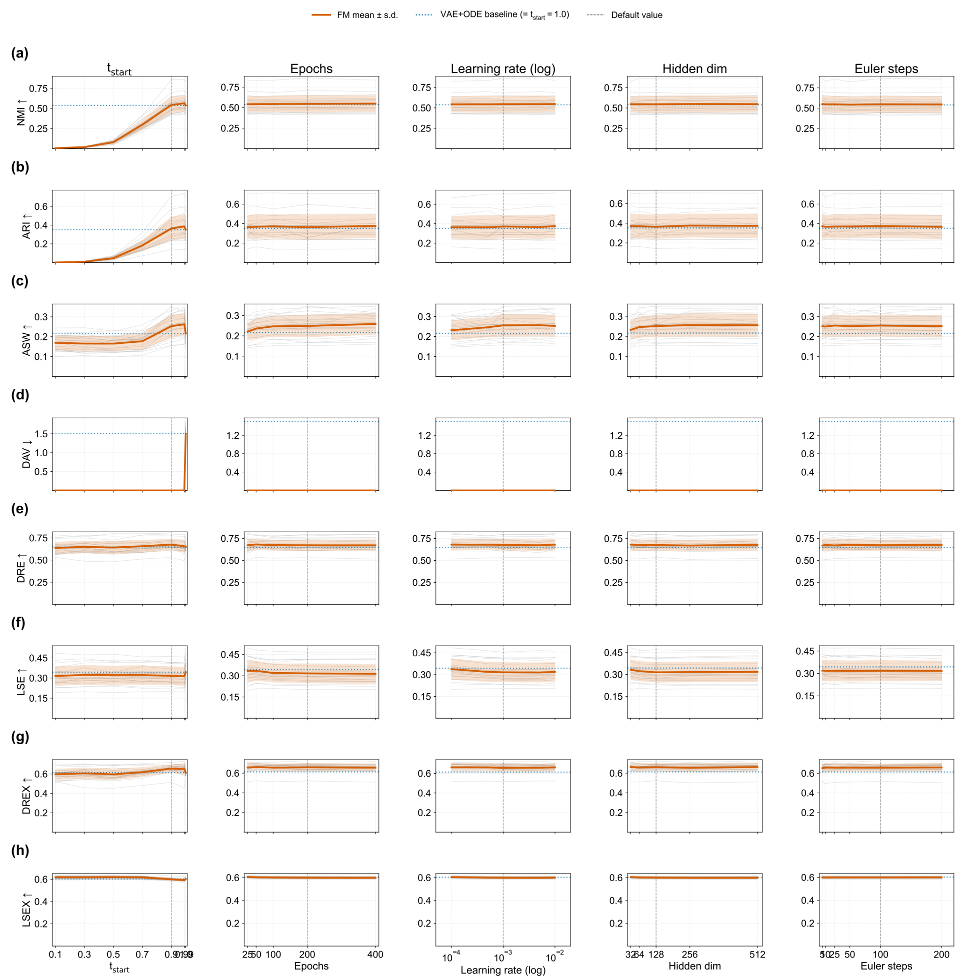

Figure A.6: FM hyperparameter sensitivity analysis across all datasets and the proposed 8-metric set (NMI, ARI, ASW, DAV, DRE, LSE, DREX, LSEX). Each row shows one metric; each column shows one hyperparameter. Per-dataset thin lines are overlaid with the mean (bold line) and ±1 s.d. shading. Vertical dashed lines mark default values; horizontal dotted lines show the VAE+ODE baseline (no FM).

Appendix C. FM Refinement Effect

Fig. C.8 quantifies the per-metric delta (FM minus base) for each of the 6 base model configurations, aggregated across all datasets. Positive values indicate FM improvement; for DAV (lower is better) the sign is flipped.

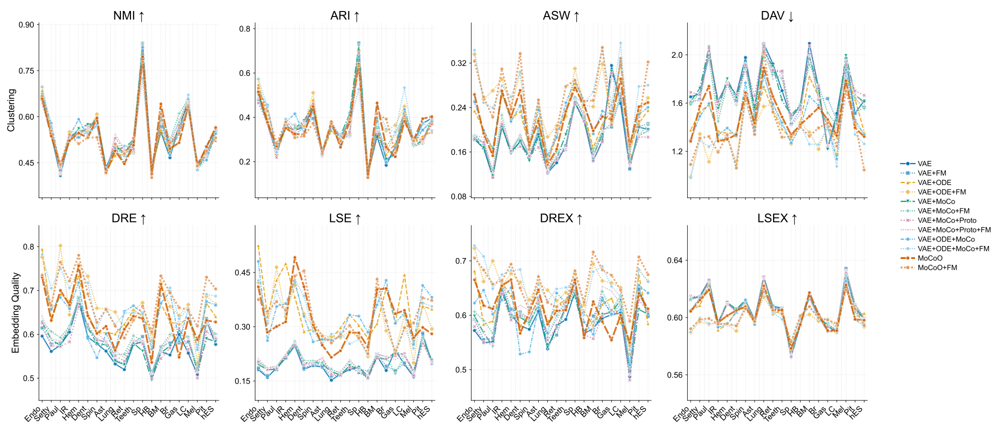

Figure B.7: FM-enhanced per-dataset metric profiles showing all 12 configurations (6 base + 6 FM variants). Each subplot shows one metric across datasets, with FM variants plotted alongside their base counterparts.

Appendix D. Per-Dataset Ablation Tables

Tables D.12–D.31 present the full per-dataset ablation results using the proposed eight-metric evaluation suite across all 20 datasets and 12 configurations (6 base + 6 FM-augmented). These complement the cross-dataset mean tables (Tables 3–4) by revealing dataset-specific strengths. FM comparison tables (Tables 7–9) appear in Section 5.6.

Appendix E. ODE-Driven Pseudotime & Trajectory Analysis

Fig. E.9 provides detailed analysis of the Neural ODE component's contribution to trajectory inference, including PCA pseudotime comparisons, pseudotime distributions per cell type, gene expression dynamics along pseudotime, and trajectory smoothness metrics.

Appendix F. Batch Integration & Cross-Dataset Generalization

Fig. F.10 provides the full cross-dataset generalization analysis referenced in Section 5.7, including column-normalised ARI heatmaps, bio-conservation vs. embedding quality scatter plots, and radar profiles.

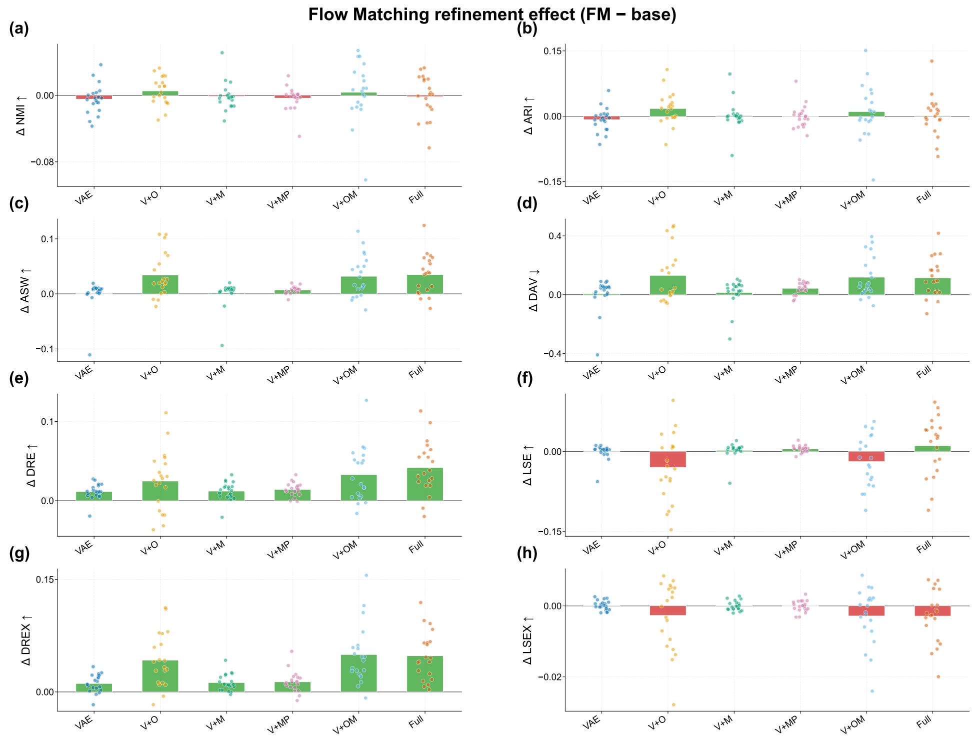

Figure C.8: FM refinement effect across model configurations. Each dot is one dataset; bars show the mean delta. Positive = FM improved the metric.

Appendix G. Biological Validation

Fig. G.11 validates the biological interpretability of MoCoO latent spaces through perturbation analysis, UMAP projections of latent dimensions and gene expression, and per-config gene–component correlation heatmaps.

Appendix H. Metric Suite Rationale

Our proposed eight-metric evaluation suite captures complementary aspects of single-cell latent-space quality. The four clustering metrics (ARI, NMI, ASW, DAV) evaluate discrete cell-type recovery at the partition level (ARI, NMI), geometric level (ASW), and compactness-to-separation ratio (DAV). The four embedding/latent-space quality metrics (DRE, LSE, DREX,

Table D.12: IRALL — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.362	0.533	0.205	1.615	0.605	0.213	0.649	0.596
VAE+ODE	0.350	0.526	0.264	1.363	0.676	**0.475**	0.660	0.591
VAE+MoCo	0.357	0.524	0.206	1.555	0.612	0.220	0.645	0.596
VAE+MoCo+Proto	0.389	0.533	0.210	1.611	0.583	0.219	0.639	0.595
VAE+ODE+MoCo	0.378	0.538	0.269	1.335	0.658	0.364	0.628	0.593
Full (MoCoO)	0.351	0.522	0.270	1.286	0.667	0.314	0.654	**0.596**
VAE+FM	0.350	0.524	0.207	1.620	0.616	0.214	0.663	0.596
VAE+ODE+FM	0.368	**0.551**	0.291	1.364	0.645	0.328	0.654	0.596
VAE+MoCo+FM	0.362	0.520	0.226	1.450	0.619	0.223	0.653	0.596
VAE+MoCo+Proto+FM	0.364	0.532	0.215	1.519	0.616	0.223	0.663	0.595
VAE+ODE+MoCo+FM	**0.392**	0.543	0.279	1.297	0.706	0.335	**0.675**	0.595
Full+FM (MoCoO+FM)	0.379	0.542	**0.309**	**1.193**	**0.722**	0.346	0.664	0.595

Table D.13: Astrocyte — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.245	**0.435**	0.192	1.550	0.562	0.188	0.607	0.599
VAE+ODE	0.237	0.421	0.221	1.426	0.604	0.265	0.630	0.595
VAE+MoCo	0.250	0.433	0.187	1.560	0.576	0.197	0.620	0.597
VAE+MoCo+Proto	0.226	0.415	0.168	1.660	0.583	0.213	0.627	0.598
VAE+ODE+MoCo	0.233	0.421	0.236	1.374	0.602	0.265	0.614	0.595
Full (MoCoO)	0.239	0.417	0.216	1.434	0.618	0.249	0.647	0.597
VAE+FM	0.246	0.433	0.202	1.500	0.582	0.192	0.631	0.598
VAE+ODE+FM	0.234	0.418	0.239	1.403	0.659	**0.276**	0.658	**0.599**
VAE+MoCo+FM	**0.251**	0.430	0.198	1.491	0.581	0.201	0.620	0.598
VAE+MoCo+Proto+FM	0.221	0.412	0.178	1.606	0.582	0.213	0.616	0.598
VAE+ODE+MoCo+FM	0.245	0.430	0.251	1.349	**0.659**	0.253	0.656	0.597
Full+FM (MoCoO+FM)	0.245	0.418	**0.254**	**1.340**	0.657	0.274	**0.671**	0.597

LSEX) each aggregate multiple sub-scores (see Appendix Appendix N for full definitions). Each composite captures a different dimension: DRE measures faithfulness of 2-D UMAP projections; DREX supplements this with rank-based and symmetry diagnostics; LSE quantifies intrinsic latent-space structure via SVD-based analysis; LSEX adds label-aware graph diagnostics. Together, these eight metrics provide a comprehensive evaluation without the redundancy of reporting 30+ individual sub-scores. Per-dataset tables

Table D.14: Brain Met. — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.314	0.551	0.144	2.095	0.560	0.218	0.576	0.616
VAE+ODE	0.333	0.557	0.167	1.819	0.670	0.308	0.582	0.614
VAE+MoCo	0.354	0.588	0.153	1.986	0.558	0.217	0.574	0.614
VAE+MoCo+Proto	0.337	0.578	0.144	2.073	0.545	0.213	0.557	0.616
VAE+ODE+MoCo	0.396	0.607	0.193	1.654	0.671	0.360	0.565	0.615
Full (MoCoO)	**0.465**	**0.642**	0.198	1.490	0.714	0.403	0.625	**0.617**
VAE+FM	0.332	0.567	0.151	2.003	0.572	0.225	0.599	0.614
VAE+ODE+FM	0.382	0.581	0.242	1.358	0.717	0.404	0.694	0.602
VAE+MoCo+FM	0.350	0.586	0.161	1.920	0.571	0.224	0.588	0.614
VAE+MoCo+Proto+FM	0.337	0.579	0.152	1.993	0.560	0.225	0.576	0.614
VAE+ODE+MoCo+FM	0.449	0.625	**0.268**	**1.260**	0.687	0.392	0.680	0.608
Full+FM (MoCoO+FM)	0.417	0.609	0.253	1.326	**0.733**	**0.432**	**0.716**	0.608

Table D.15: Breast — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.184	0.467	0.180	1.724	0.553	0.179	0.594	0.601
VAE+ODE	0.233	0.478	0.210	1.653	0.628	0.407	0.620	0.602
VAE+MoCo	0.212	0.507	0.191	1.725	0.574	0.209	0.602	0.601
VAE+MoCo+Proto	0.255	0.503	0.178	1.743	0.569	0.211	0.599	**0.602**
VAE+ODE+MoCo	0.197	0.476	0.233	1.582	0.697	**0.428**	0.632	0.601
Full (MoCoO)	0.265	0.496	0.224	1.562	0.635	0.406	0.588	0.602
VAE+FM	0.212	0.491	0.200	1.633	0.580	0.191	0.628	0.599
VAE+ODE+FM	0.340	0.511	0.318	**1.218**	0.663	0.289	0.663	0.598
VAE+MoCo+FM	0.266	**0.525**	0.198	1.748	0.599	0.217	0.644	0.599
VAE+MoCo+Proto+FM	0.248	0.509	0.198	1.641	0.595	0.218	0.653	0.599
VAE+ODE+MoCo+FM	0.347	0.522	0.347	1.257	**0.706**	0.318	**0.690**	0.599
Full+FM (MoCoO+FM)	**0.392**	0.516	**0.348**	1.395	0.695	0.316	0.683	0.600

for all 20 datasets appear in Appendix Appendix D.

Appendix I. Multi-Seed Robustness

To assess the stability of MoCoO across random initialisations, we train all base configurations with 5 independent seeds on IRALL (β=0.1, 200 epochs, 3,000 cells). Fig. I.12 shows per-seed score distributions (box + strip plots) and mean ARI with bootstrapped 95% confidence intervals. ODE-containing

Table D.16: Dentate — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.360	0.554	0.179	1.671	0.594	0.183	0.581	0.605
VAE+ODE	0.362	0.562	0.233	1.355	0.647	0.287	0.581	0.604
VAE+MoCo	**0.385**	**0.575**	0.179	1.644	0.614	0.198	0.596	**0.606**
VAE+MoCo+Proto	0.381	0.566	0.170	1.672	0.622	0.195	0.600	0.605
VAE+ODE+MoCo	0.327	0.537	0.243	1.327	0.591	0.329	0.529	0.604
Full (MoCoO)	0.322	0.531	0.271	1.339	0.642	0.411	0.567	0.605
VAE+FM	0.356	0.550	0.188	1.623	0.615	0.186	0.606	0.604
VAE+ODE+FM	0.363	0.552	0.303	1.119	**0.732**	0.335	0.661	0.591
VAE+MoCo+FM	0.375	0.562	0.190	1.584	0.617	0.207	0.603	0.604
VAE+MoCo+Proto+FM	0.374	0.564	0.188	1.592	0.642	0.200	0.618	0.604
VAE+ODE+MoCo+FM	0.338	0.546	0.304	1.076	0.717	0.385	**0.684**	0.594
Full+FM (MoCoO+FM)	0.315	0.532	**0.337**	**1.061**	0.692	**0.455**	0.650	0.594

Table D.17: Endo — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.512	0.665	0.183	1.652	0.596	0.181	0.572	0.613
VAE+ODE	0.547	0.685	0.234	1.376	**0.794**	**0.524**	0.681	0.605
VAE+MoCo	0.502	0.667	0.186	1.612	0.628	0.196	0.598	0.614
VAE+MoCo+Proto	0.465	0.643	0.183	1.689	0.626	0.203	0.579	**0.615**
VAE+ODE+MoCo	0.512	0.674	0.250	1.294	0.731	0.481	0.622	0.605
Full (MoCoO)	0.515	0.659	0.264	1.283	0.731	0.411	0.665	0.604
VAE+FM	0.482	0.656	0.189	1.604	0.615	0.186	0.592	0.613
VAE+ODE+FM	0.571	0.696	0.336	0.989	0.776	0.422	0.723	0.590
VAE+MoCo+FM	0.488	0.654	0.190	1.589	0.630	0.203	0.606	0.613
VAE+MoCo+Proto+FM	0.487	0.644	0.190	1.624	0.642	0.211	0.592	0.614
VAE+ODE+MoCo+FM	**0.573**	**0.697**	**0.343**	**0.983**	0.782	0.428	**0.728**	0.590
Full+FM (MoCoO+FM)	0.532	0.668	0.324	1.091	0.735	0.375	0.705	0.592

configurations exhibit moderate seed-to-seed variability in ARI but consistently outperform non-ODE variants.

Appendix J. External Baseline Comparison

We compare MoCoO against five external baselines: scVI [5], DPT [12], PCA+KMeans, Harmony [13], and scANVI [14] across seven datasets (IRALL, paul, endo, dentate, hemato, lung, spinoids) with five random seeds each.

Table D.18: Gastric — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.269	0.575	**0.316**	1.228	0.600	0.235	0.601	0.594
VAE+ODE	0.282	0.573	0.235	1.392	0.623	0.311	0.631	0.591
VAE+MoCo	0.234	0.559	0.305	**1.217**	0.609	0.230	0.605	0.594
VAE+MoCo+Proto	0.238	0.563	0.302	1.247	0.614	0.220	0.609	0.594
VAE+ODE+MoCo	0.263	0.554	0.228	1.406	0.584	0.309	0.602	0.590
Full (MoCoO)	0.222	0.515	0.218	1.464	0.548	**0.336**	0.554	0.591
VAE+FM	0.227	0.543	0.205	1.637	0.581	0.178	0.602	0.596
VAE+ODE+FM	**0.365**	0.588	0.212	1.427	0.643	0.233	0.660	**0.599**
VAE+MoCo+FM	0.331	**0.610**	0.211	1.518	0.588	0.171	0.608	0.595
VAE+MoCo+Proto+FM	0.319	0.587	0.291	1.253	0.634	0.210	0.645	0.590
VAE+ODE+MoCo+FM	0.334	0.582	0.244	1.397	0.636	0.229	0.653	0.599
Full+FM (MoCoO+FM)	0.240	0.547	0.209	1.435	**0.661**	0.226	**0.674**	0.598

Table D.19: Hemato — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.339	0.554	0.160	1.768	0.677	0.247	0.601	0.610
VAE+ODE	0.347	0.551	0.220	1.350	0.733	0.383	0.636	0.595
VAE+MoCo	0.350	0.561	0.157	1.801	0.677	0.256	0.594	**0.611**
VAE+MoCo+Proto	0.334	0.542	0.162	1.764	0.663	0.250	0.591	0.610
VAE+ODE+MoCo	**0.416**	**0.592**	0.215	1.394	0.718	0.425	0.651	0.595
Full (MoCoO)	0.325	0.547	0.223	**1.300**	0.756	**0.492**	0.666	0.601
VAE+FM	0.320	0.534	0.161	1.761	0.683	0.252	0.611	0.610
VAE+ODE+FM	0.365	0.549	0.239	1.314	0.759	0.415	0.677	0.595
VAE+MoCo+FM	0.342	0.553	0.161	1.757	0.683	0.260	0.597	0.610
VAE+MoCo+Proto+FM	0.337	0.542	0.167	1.769	0.675	0.255	0.604	0.610
VAE+ODE+MoCo+FM	0.361	0.550	**0.259**	1.317	0.746	0.430	0.682	0.595
Full+FM (MoCoO+FM)	0.308	0.512	0.237	1.336	**0.780**	0.440	**0.694**	0.596

Fig. J.13 shows grouped bar charts of ARI/NMI/ASW and a normalised ranking. MoCoO's Full model achieves the highest composite normalised score across all methods. Note that Harmony is limited to datasets with batch annotations; scANVI uses cell-type labels in a semi-supervised manner.

Table D.20: Hepatoblastoma — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.144	0.433	0.217	1.578	0.505	0.164	0.566	0.594
VAE+ODE	0.135	0.420	0.232	1.434	0.603	0.268	0.607	0.594
VAE+MoCo	0.155	0.428	0.214	1.585	0.495	0.157	0.567	0.595
VAE+MoCo+Proto	0.148	**0.444**	0.216	1.544	0.499	0.161	0.564	0.595
VAE+ODE+MoCo	0.153	0.427	0.250	**1.400**	0.567	**0.285**	0.587	0.595
Full (MoCoO)	0.128	0.413	0.244	1.419	0.536	0.244	0.558	0.595
VAE+FM	0.150	0.436	0.227	1.546	0.512	0.158	0.571	0.596
VAE+ODE+FM	**0.161**	0.436	0.244	1.440	0.585	0.218	0.619	0.600
VAE+MoCo+FM	0.155	0.429	0.222	1.583	0.510	0.178	0.567	0.596
VAE+MoCo+Proto+FM	0.160	0.441	0.228	1.505	0.511	0.167	0.567	0.596
VAE+ODE+MoCo+FM	0.141	0.416	0.247	1.419	0.572	0.222	0.600	0.600
Full+FM (MoCoO+FM)	0.128	0.402	**0.251**	1.403	**0.606**	0.226	**0.624**	**0.601**

Table D.21: hESC-time — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.396	0.557	0.201	1.619	0.577	0.199	0.601	0.599
VAE+ODE	0.334	0.523	0.235	1.361	0.640	0.349	0.585	**0.603**
VAE+MoCo	0.391	0.555	0.198	1.608	0.583	0.208	0.602	0.599
VAE+MoCo+Proto	0.395	0.561	0.187	1.664	0.591	0.202	0.607	0.599
VAE+ODE+MoCo	0.381	0.545	0.240	1.339	0.666	**0.381**	0.641	0.594
Full (MoCoO)	**0.403**	**0.565**	0.249	1.322	0.628	0.280	0.612	0.598
VAE+FM	0.387	0.551	0.211	1.567	0.589	0.203	0.598	0.600
VAE+ODE+FM	0.379	0.545	0.257	1.344	0.669	0.294	0.662	0.594
VAE+MoCo+FM	0.392	0.553	0.209	1.554	0.601	0.210	0.611	0.599
VAE+MoCo+Proto+FM	0.376	0.546	0.197	1.589	0.595	0.201	0.613	0.599
VAE+ODE+MoCo+FM	0.340	0.528	0.280	1.262	0.687	0.356	0.662	0.596
Full+FM (MoCoO+FM)	0.370	0.532	**0.322**	**1.046**	**0.704**	0.373	**0.675**	0.595

Appendix K. Trajectory and Pseudotime Comparison

Fig. K.14 evaluates the ODE-derived pseudotime against dedicated trajectory inference methods. Panel (a) shows the mean absolute Spearman correlation between predicted pseudotime and collection day for each ODE-containing configuration. Panel (b) compares MoCoO configurations against DPT and Palantir on the same task.

Table D.22: Liver Cancer — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.399	0.640	0.247	1.515	0.557	0.196	0.605	0.590
VAE+ODE	0.428	0.646	0.319	1.321	0.636	**0.444**	0.587	0.590
VAE+MoCo	0.411	0.656	0.256	1.473	0.590	0.222	0.628	0.590
VAE+MoCo+Proto	0.397	0.647	0.261	1.429	0.566	0.226	0.600	0.590
VAE+ODE+MoCo	0.436	0.624	0.325	1.138	0.657	0.339	0.610	0.592
Full (MoCoO)	0.382	0.635	0.292	1.331	0.637	0.345	0.600	0.591
VAE+FM	0.395	0.636	0.259	1.457	0.583	0.200	0.620	0.591
VAE+ODE+FM	0.449	0.650	0.319	1.173	**0.668**	0.331	**0.651**	0.594
VAE+MoCo+FM	0.413	0.656	0.265	1.418	0.594	0.219	0.637	0.591
VAE+MoCo+Proto+FM	0.403	0.647	0.273	1.397	0.590	0.221	0.620	0.592
VAE+ODE+MoCo+FM	**0.534**	**0.671**	**0.356**	**1.075**	0.660	0.297	0.634	0.592
Full+FM (MoCoO+FM)	0.405	0.631	0.329	1.244	0.656	0.297	0.646	**0.596**

Table D.23: Lung — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.357	0.505	0.122	2.095	0.532	0.152	0.538	0.628
VAE+ODE	0.341	0.475	0.144	1.738	0.601	0.265	0.587	0.621
VAE+MoCo	0.355	0.494	0.124	2.029	0.539	0.175	0.541	**0.628**
VAE+MoCo+Proto	0.375	0.530	0.125	2.064	0.551	0.170	0.554	0.628
VAE+ODE+MoCo	0.362	0.499	0.155	1.827	0.561	0.262	0.567	0.621
Full (MoCoO)	0.380	0.489	0.137	1.893	0.564	0.215	0.583	0.622
VAE+FM	0.358	0.506	0.124	2.075	0.541	0.163	0.546	0.628
VAE+ODE+FM	0.378	0.504	**0.187**	**1.572**	**0.651**	0.272	**0.646**	0.614
VAE+MoCo+FM	0.356	0.494	0.128	2.006	0.553	0.183	0.560	0.627
VAE+MoCo+Proto+FM	**0.385**	**0.542**	0.127	2.098	0.563	0.175	0.563	0.627
VAE+ODE+MoCo+FM	0.359	0.505	0.148	1.860	0.622	**0.280**	0.626	0.618
Full+FM (MoCoO+FM)	0.372	0.511	0.137	2.023	0.619	0.255	0.624	0.620

Appendix L. Beta Ablation Heatmap

Fig. L.15 provides a compact heatmap view of the beta sensitivity analysis, showing ARI, NMI, and ASW for all six configurations at $\beta \in \{0.01, 0.1, 1.0\}$. This complements the line-plot view in Fig. A.6.

Table D.24: Melanoma — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.288	0.433	0.130	1.951	0.508	0.162	0.487	0.632
VAE+ODE	**0.302**	0.443	0.165	1.830	0.571	0.254	0.545	0.624
VAE+MoCo	0.271	0.433	0.141	1.995	0.515	0.178	0.490	0.630
VAE+MoCo+Proto	0.296	0.450	0.136	1.917	0.500	0.165	0.481	0.632
VAE+ODE+MoCo	0.299	**0.454**	0.172	1.878	0.565	0.251	0.508	0.626
Full (MoCoO)	0.297	0.441	0.171	1.787	0.588	0.270	0.550	0.623
VAE+FM	0.279	0.428	0.130	1.967	0.520	0.165	0.497	**0.634**
VAE+ODE+FM	0.291	0.436	0.155	1.873	0.535	0.200	0.528	0.630
VAE+MoCo+FM	0.270	0.429	0.144	1.927	0.524	0.175	0.503	0.633
VAE+MoCo+Proto+FM	0.267	0.434	0.136	1.958	0.514	0.176	0.491	0.633
VAE+ODE+MoCo+FM	0.290	0.438	0.161	1.823	0.570	0.240	0.536	0.629
Full+FM (MoCoO+FM)	0.294	0.440	**0.173**	**1.700**	**0.619**	**0.310**	**0.589**	0.619

Table D.25: Paul — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.245	0.408	0.115	2.016	0.575	0.187	0.551	0.626
VAE+ODE	0.235	0.438	0.163	1.585	0.691	0.429	0.589	0.623
VAE+MoCo	0.243	0.423	0.118	2.063	0.575	0.183	0.543	0.626
VAE+MoCo+Proto	0.219	0.417	0.114	2.057	0.572	0.186	0.546	**0.626**
VAE+ODE+MoCo	0.244	0.437	0.167	1.596	0.684	0.356	0.602	0.620
Full (MoCoO)	0.269	0.441	0.153	1.741	0.700	0.301	0.612	0.619
VAE+FM	0.250	0.414	0.119	2.004	0.581	0.181	0.558	0.624
VAE+ODE+FM	0.232	0.430	**0.270**	**1.114**	**0.802**	**0.465**	**0.700**	0.595
VAE+MoCo+FM	0.241	0.417	0.128	1.969	0.592	0.190	0.567	0.624
VAE+MoCo+Proto+FM	0.253	0.416	0.122	2.017	0.585	0.191	0.547	0.625
VAE+ODE+MoCo+FM	0.242	0.436	0.239	1.239	0.750	0.399	0.649	0.596
Full+FM (MoCoO+FM)	**0.282**	**0.445**	0.223	1.323	0.765	0.370	0.657	0.599

Table D.26: Pituitary — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.373	**0.503**	0.205	1.516	0.630	0.279	0.649	0.607
VAE+ODE	0.315	0.495	0.211	1.631	0.671	0.356	0.674	0.601
VAE+MoCo	0.295	0.475	0.190	1.668	0.590	0.263	0.611	0.606
VAE+MoCo+Proto	0.288	0.472	0.187	1.702	0.606	0.254	0.628	0.606
VAE+ODE+MoCo	0.343	0.459	0.228	1.459	0.675	**0.414**	0.655	0.599
Full (MoCoO)	**0.394**	0.503	0.242	1.395	0.632	0.298	0.640	0.599
VAE+FM	0.342	0.485	0.206	1.516	0.636	0.282	0.648	0.609
VAE+ODE+FM	0.286	0.471	0.231	1.430	0.688	0.365	0.683	0.608
VAE+MoCo+FM	0.282	0.462	0.191	1.643	0.623	0.269	0.636	0.609
VAE+MoCo+Proto+FM	0.288	0.474	0.190	1.675	0.625	0.260	0.639	**0.610**
VAE+ODE+MoCo+FM	0.375	0.497	**0.278**	**1.340**	0.692	0.366	0.684	0.604
Full+FM (MoCoO+FM)	0.302	0.489	0.215	1.442	**0.730**	0.380	**0.707**	0.606

Table D.27: Retina — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.311	0.483	0.141	1.926	0.520	0.169	0.581	0.608
VAE+ODE	0.294	0.477	0.187	1.537	0.623	**0.298**	0.617	0.603
VAE+MoCo	0.318	0.499	0.149	1.906	0.531	0.167	0.576	0.607
VAE+MoCo+Proto	0.304	0.492	0.146	1.874	0.539	0.165	0.578	0.606
VAE+ODE+MoCo	0.275	0.451	0.186	1.592	**0.652**	0.274	0.641	0.606
Full (MoCoO)	0.264	0.446	0.165	1.655	0.612	0.234	0.608	0.605
VAE+FM	0.302	0.473	0.150	1.864	0.533	0.167	0.564	**0.608**
VAE+ODE+FM	0.305	0.487	**0.207**	1.596	0.610	0.270	0.627	0.604
VAE+MoCo+FM	**0.323**	0.481	0.160	1.836	0.555	0.172	0.599	0.607
VAE+MoCo+Proto+FM	0.309	0.495	0.152	1.797	0.547	0.176	0.592	0.606
VAE+ODE+MoCo+FM	0.316	**0.505**	0.195	1.557	0.648	0.284	**0.648**	0.603
Full+FM (MoCoO+FM)	0.293	0.480	0.200	**1.526**	0.592	0.279	0.615	0.605

Table D.28: Setty — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.371	0.541	0.167	1.677	0.561	0.160	0.550	**0.615**
VAE+ODE	0.395	0.554	0.200	1.528	0.635	0.329	0.583	0.608
VAE+MoCo	0.367	0.537	0.168	1.723	0.588	0.181	0.568	0.614
VAE+MoCo+Proto	0.369	0.546	0.178	1.670	0.574	0.181	0.551	0.614
VAE+ODE+MoCo	**0.455**	**0.579**	0.199	1.490	0.657	0.262	0.646	0.614
Full (MoCoO)	0.378	0.537	0.193	1.575	0.632	0.285	0.618	0.612
VAE+FM	0.375	0.543	0.175	1.631	0.578	0.164	0.555	0.615
VAE+ODE+FM	0.391	0.553	0.255	1.344	0.657	0.311	0.662	0.597
VAE+MoCo+FM	0.382	0.544	0.176	1.690	0.606	0.186	0.590	0.614
VAE+MoCo+Proto+FM	0.370	0.545	0.186	1.619	0.589	0.187	0.567	0.613
VAE+ODE+MoCo+FM	0.447	0.570	0.249	**1.291**	**0.705**	0.309	**0.708**	0.600
Full+FM (MoCoO+FM)	0.428	0.554	**0.259**	1.309	0.666	**0.343**	0.683	0.599

Table D.29: Spine — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.677	0.801	0.248	1.477	0.583	0.188	0.643	0.574
VAE+ODE	0.703	0.834	0.283	1.308	0.642	0.304	0.646	0.577
VAE+MoCo	0.725	**0.840**	0.249	1.500	0.562	0.183	0.635	0.572
VAE+MoCo+Proto	0.692	0.833	0.246	1.496	0.578	0.178	0.655	0.573
VAE+ODE+MoCo	0.673	0.826	0.288	1.271	0.649	**0.329**	0.666	0.579
Full (MoCoO)	0.639	0.796	0.276	1.336	0.636	0.283	0.665	0.578
VAE+FM	**0.736**	0.838	0.257	1.424	0.595	0.173	0.646	0.575
VAE+ODE+FM	0.701	0.826	**0.310**	**1.262**	**0.672**	0.271	0.676	0.582
VAE+MoCo+FM	0.726	0.839	0.256	1.474	0.586	0.184	0.646	0.574
VAE+MoCo+Proto+FM	0.647	0.783	0.254	1.413	0.586	0.178	0.650	0.574
VAE+ODE+MoCo+FM	0.526	0.725	0.259	1.345	0.646	0.265	0.658	0.581
Full+FM (MoCoO+FM)	0.631	0.779	0.289	1.321	0.664	0.269	**0.682**	**0.583**

Table D.30: Spinoids — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.442	0.598	0.150	1.978	0.578	0.193	0.574	0.612
VAE+ODE	0.480	0.598	0.168	1.696	0.603	0.293	0.596	0.607
VAE+MoCo	0.452	0.582	0.144	1.923	0.579	0.202	0.586	**0.612**
VAE+MoCo+Proto	0.462	0.587	0.153	1.916	0.587	0.200	0.595	0.612
VAE+ODE+MoCo	0.430	0.574	0.171	1.768	0.547	0.258	0.533	0.612
Full (MoCoO)	0.449	0.596	0.173	1.687	0.602	0.306	0.606	0.608
VAE+FM	0.395	0.572	0.152	1.893	0.589	0.197	0.593	0.611
VAE+ODE+FM	**0.511**	**0.608**	0.171	1.682	**0.660**	0.300	**0.629**	0.610
VAE+MoCo+FM	0.447	0.598	0.154	1.924	0.588	0.206	0.582	0.612
VAE+MoCo+Proto+FM	0.444	0.580	0.159	1.882	0.604	0.201	0.609	0.612
VAE+ODE+MoCo+FM	0.421	0.574	**0.184**	**1.622**	0.614	0.288	0.613	0.608
Full+FM (MoCoO+FM)	0.374	0.533	0.165	1.662	0.627	**0.313**	0.609	0.606

Table D.31: Teeth — proposed eight-metric evaluation suite (split=whole, all 12 configurations). Best per column in **bold**. DAV↓ indicates lower is better.

Config	ARI	NMI	ASW	DAV↓	DRE overall	LSE overall	DREX overall	LSEX overall
VAE	0.385	0.518	0.171	1.703	0.577	0.182	0.592	**0.607**
VAE+ODE	0.390	0.514	0.236	1.386	0.655	**0.333**	0.629	0.606
VAE+MoCo	**0.416**	0.524	0.191	1.623	0.580	0.192	0.616	0.606
VAE+MoCo+Proto	0.335	0.495	0.165	1.867	0.585	0.187	0.617	0.605
VAE+ODE+MoCo	0.411	0.519	0.249	1.437	**0.665**	0.333	0.623	0.605
Full (MoCoO)	0.380	0.507	0.233	1.486	0.642	0.284	0.610	0.605
VAE+FM	0.321	0.481	0.164	1.859	0.581	0.189	0.615	0.606
VAE+ODE+FM	0.325	0.484	0.224	1.435	0.655	0.286	0.642	0.602
VAE+MoCo+FM	0.326	0.493	0.169	1.806	0.597	0.205	0.626	0.606
VAE+MoCo+Proto+FM	0.310	0.480	0.166	1.841	0.585	0.209	0.623	0.606
VAE+ODE+MoCo+FM	0.371	0.505	0.243	1.415	0.649	0.253	**0.651**	0.599
Full+FM (MoCoO+FM)	0.393	**0.530**	**0.278**	**1.317**	0.632	0.303	0.624	0.603

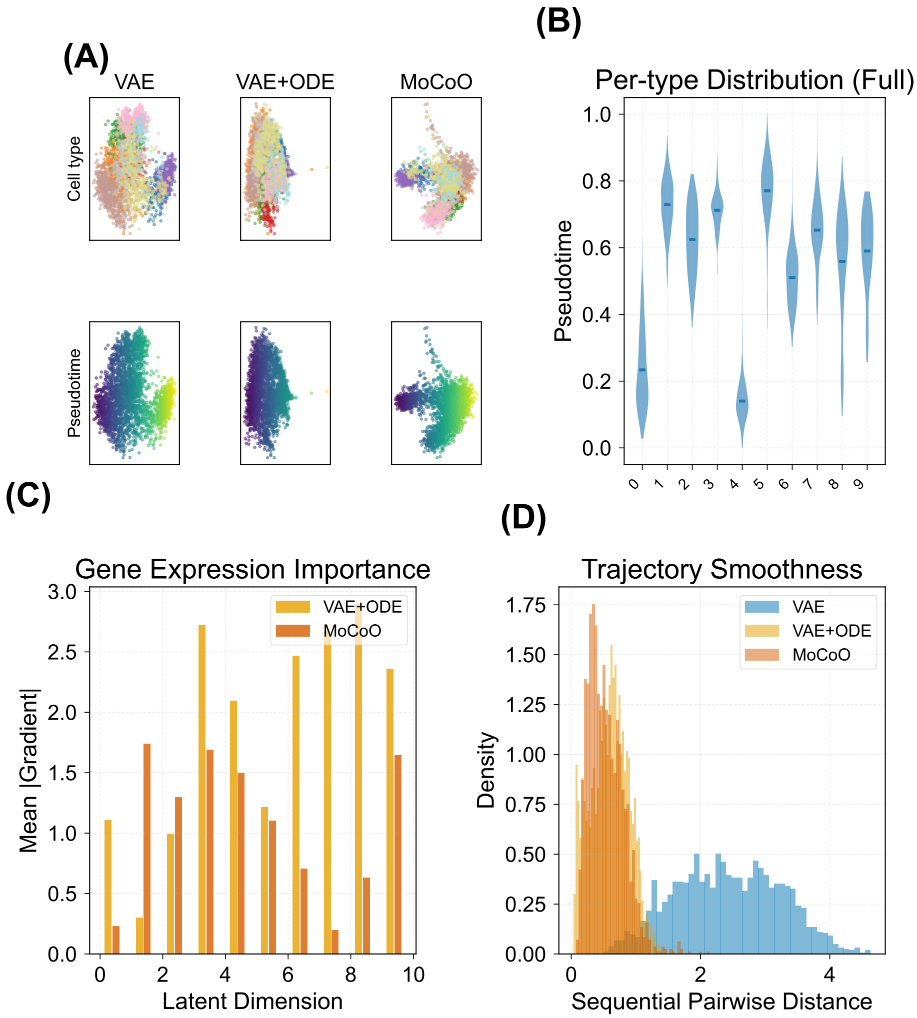

Figure E.9: ODE-driven pseudotime and trajectory analysis (IRALL). **(A)** PCA of latent spaces coloured by cell type (top row) and pseudotime (bottom row) for three representative configurations (VAE, VAE+ODE, Full); ODE adds temporal directionality. **(B)** Per-cell-type pseudotime distributions (violin plots, Full model); the ODE model separates developmental stages more continuously. **(C)** Per-dimension gradient magnitude across configurations, quantifying gene expression importance along the ODE axis. **(D)** Trajectory smoothness: sequential pairwise distance histograms per configuration.

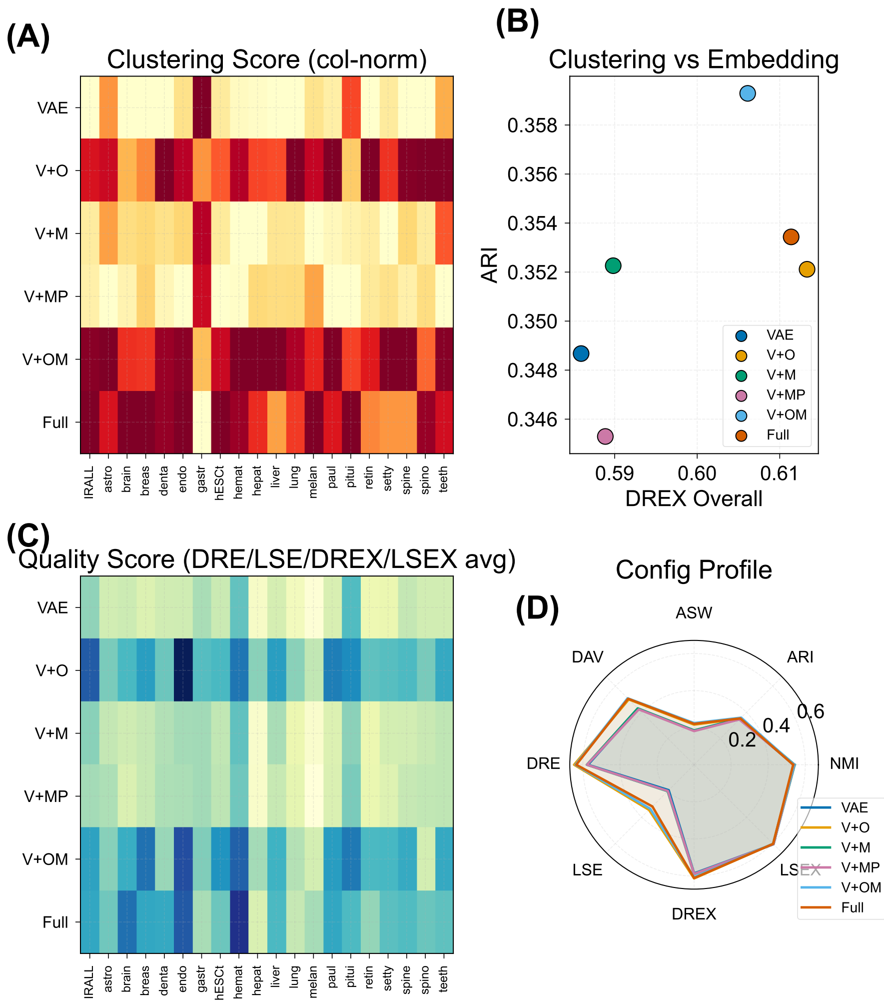

Figure F.10: Cross-dataset evaluation and generalization. **(A)** Column-normalised ARI heatmap (config × dataset) showing relative performance across all 20 datasets. **(B)** Bio-conservation (NMI) vs. embedding quality (DREX overall) scatter; each point is one configuration averaged across datasets. **(C)** Raw NMI heatmap across all datasets. **(D)** Radar profile of normalised configuration performance (ARI, NMI, ASW) averaged across datasets.

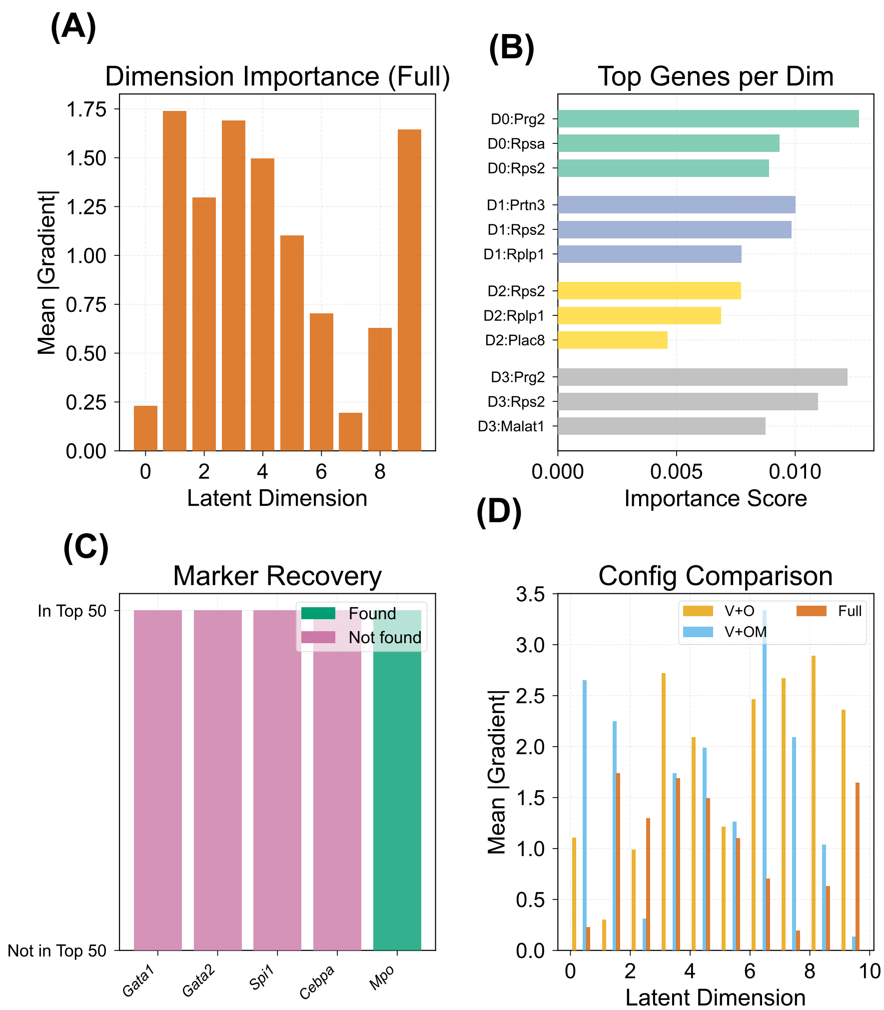

Figure G.11: Biological validation of MoCoO latent spaces (IRALL). **(A)** Per-dimension gradient importance for the Full model, quantifying the contribution of each latent dimension. **(B)** Top genes per latent component ranked by Random-Forest importance score. **(C)** Marker gene recovery: known marker genes checked against the top-50 most important genes. **(D)** Cross-configuration gradient importance comparison for all six base configurations.

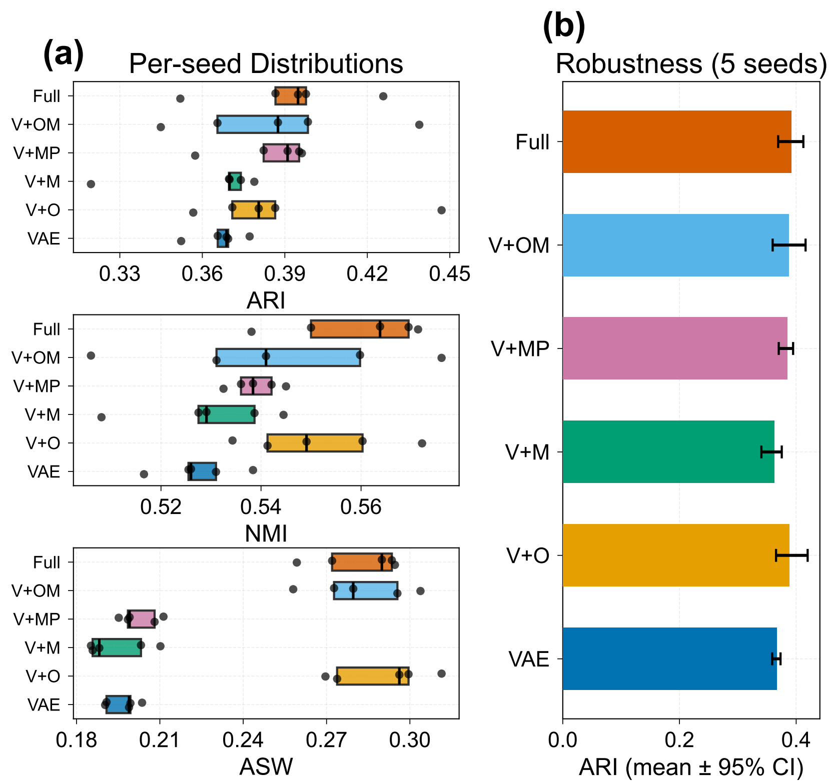

Figure I.12: Multi-seed robustness (IRALL, 5 seeds, β=0.1). **(a)** Per-seed ARI, NMI, and ASW distributions across six configurations (box + strip plots; whiskers omitted for clarity, individual seeds shown as points). **(b)** Mean ARI with 95% confidence intervals per configuration.

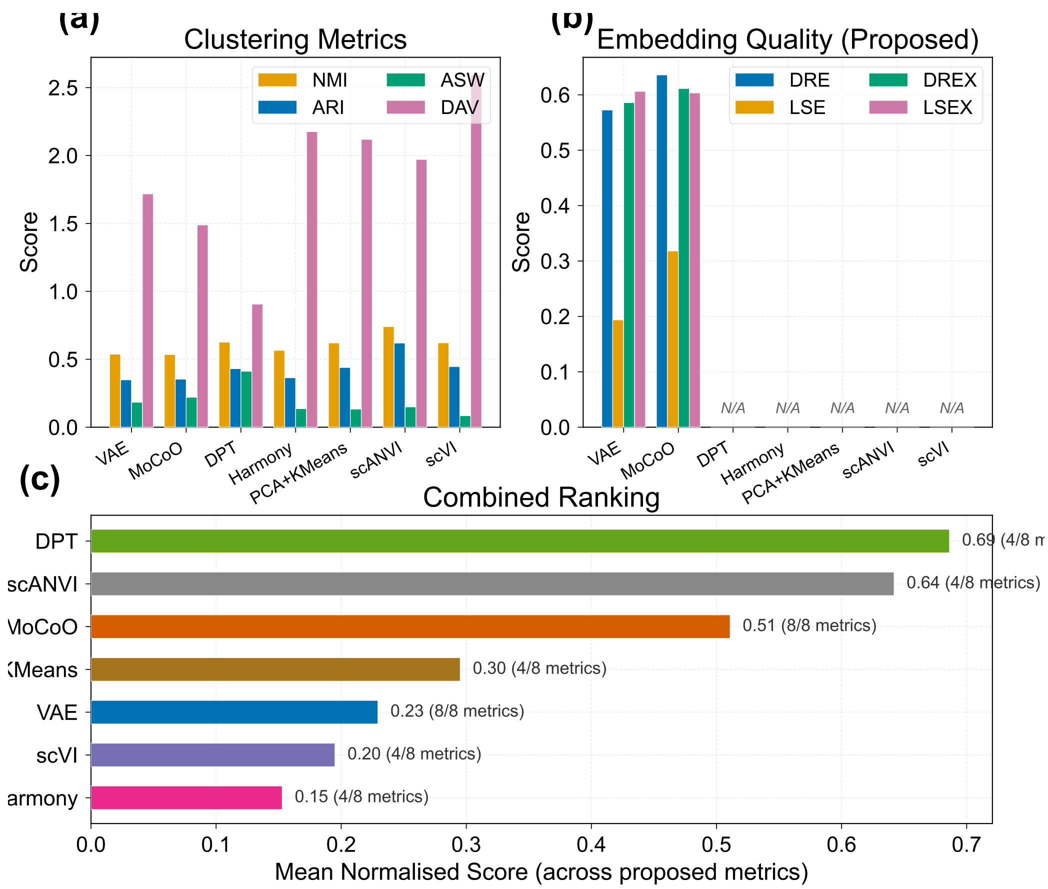

Figure J.13: External baseline comparison. **(a)** Grouped bar chart of ARI, NMI, and ASW for MoCoO configurations and external methods. **(b)** Normalised composite ranking (min–max per metric, then averaged).

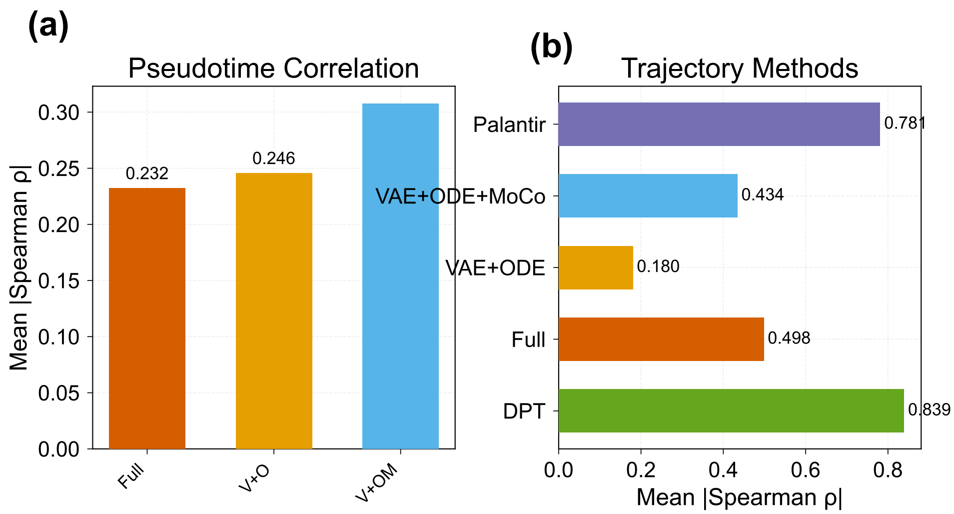

Figure K.14: Trajectory and pseudotime comparison. **(a)** Mean |Spearman ρ| between inferred pseudotime and collection day per ODE-containing configuration. **(b)** Trajectory baseline comparison: MoCoO vs. DPT and Palantir.

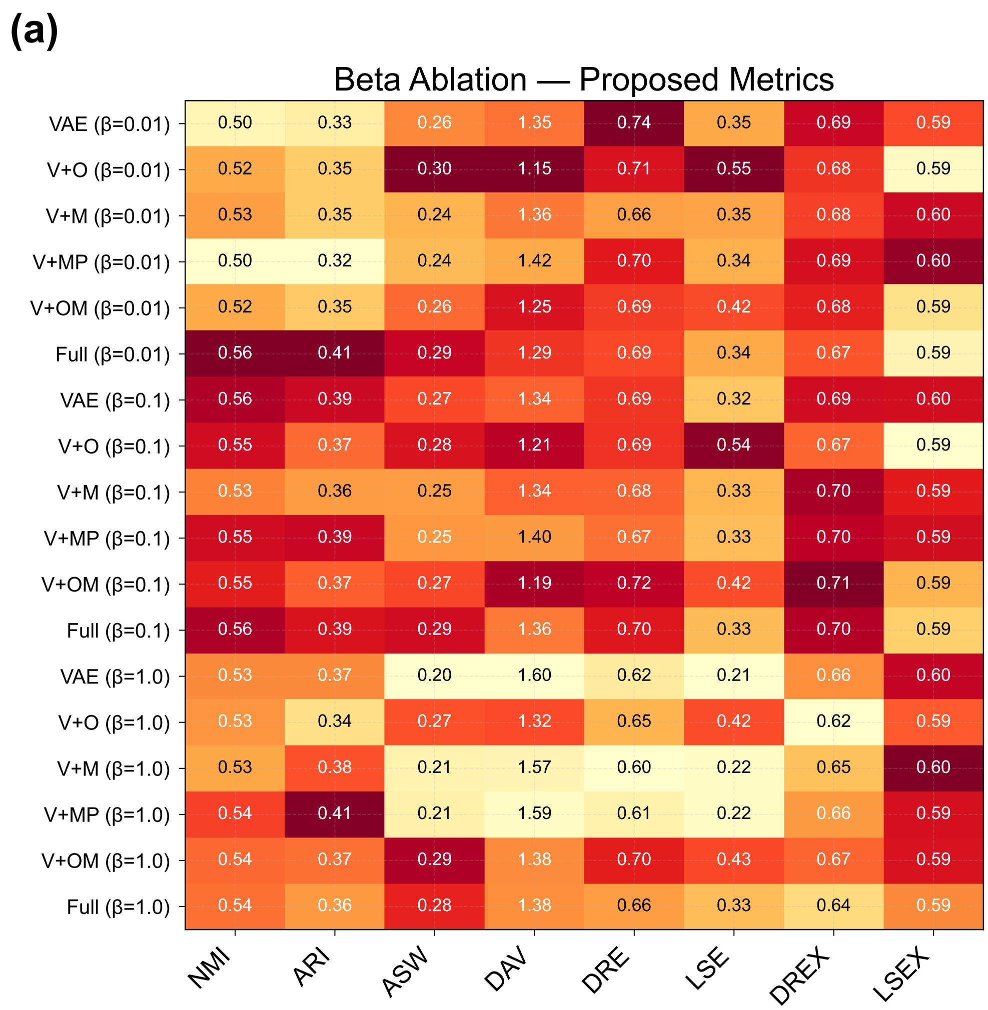

Figure L.15: Beta ablation heatmap. Each cell shows metric × configuration at a given β value, with text annotations and colour-mapped cells. Contrast-aware text colour ensures readability.

Table N.32: IRALL pseudotime–marker correlations (hematopoiesis).

Gene	Function	Spearman ρ	p-value
Hbb-bs	Erythroid hemoglobin	+0.275	$< 10^{-53}$
Hba-a1	Erythroid hemoglobin	+0.264	$< 10^{-49}$
Elane	Granulocyte elastase	+0.255	$< 10^{-45}$
Cd34	HSC / progenitor	−0.233	$< 10^{-38}$
Ctsg	Granulocyte cathepsin G	+0.226	$< 10^{-36}$
Cd8a	T-cell marker	−0.168	$< 10^{-20}$

Appendix M. Pseudotime–Marker Correlation Tables

Tables N.32–N.36 present pseudotime–marker correlations for each dataset (Full MoCoO, 200 epochs). See Section 5.3 for the gene selection protocol and Table 5 for the cross-dataset summary.

Appendix N. LSE/LSEX Metric Definitions

LSE assesses the intrinsic quality of the latent space. Let $\sigma_1 \geq \cdots \geq \sigma_d$ denote the singular values of the mean-centred latent matrix $Z \in \mathbb{R}^{N \times d}$. (i) *Manifold dimensionality*: $\mathrm{PR} = (\sum_j \sigma_j^2)^2 / \sum_j \sigma_j^4$, normalised by d. (ii) *Spectral decay rate*: slope of $\log \sigma_j$ versus j. (iii) *Anisotropy score*: $\sigma_1/\bar{\sigma}$. (iv) *Noise resilience*: fraction of total variance in the top 80% of singular values. *LSE overall quality*: arithmetic mean of the above.

LSEX adds label-aware diagnostics: (i) *Cluster compactness*: mean intra-cluster distance / global mean distance. (ii) *Inter-cluster gap*: mean nearest-centroid distance between clusters. (iii) *k-NN purity*: fraction of k-nearest neighbours sharing the same label. (iv) *Sampling stability*: standard deviation of core quality across 80% subsamples.

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Table N.33: Dentate pseudotime–marker correlations (neurogenesis).

Gene	Function	Spearman ρ	p-value
Slc1a3	Astrocyte/RGC (GLAST)	+0.542	$< 10^{-228}$
Fabp7	Astrocyte/RGC (BLBP)	+0.476	$< 10^{-169}$
Sox2	Neural stem cell	+0.439	$< 10^{-141}$
Gfap	Astrocyte (GFAP)	+0.376	$< 10^{-101}$
Olig1	Oligodendrocyte	+0.282	$< 10^{-56}$
Dcx	Migrating neuroblast	−0.229	$< 10^{-37}$

Table N.34: Endo pseudotime–marker correlations (pancreatic endocrine).

Gene	Function	Spearman ρ	p-value
Ins2	β-cell insulin	+0.463	$< 10^{-134}$
Ins1	β-cell insulin	+0.372	$< 10^{-83}$
Neurog3	Endocrine progenitor	−0.319	$< 10^{-61}$
Pdx1	β-cell / progenitor TF	+0.280	$< 10^{-47}$
Chgb	Pan-endocrine	+0.191	$< 10^{-22}$
Gcg	α-cell glucagon	+0.181	$< 10^{-20}$

Table N.35: Paul pseudotime–marker correlations (myeloid/erythroid).

Gene	Function	Spearman ρ	p-value
Epor	Erythropoietin receptor	+0.346	$< 10^{-77}$
Hba-a2	Erythroid hemoglobin	+0.328	$< 10^{-69}$
Ly6c2	Myeloid surface marker	+0.272	$< 10^{-47}$
Elane	Granulocyte elastase	+0.253	$< 10^{-41}$
Gata1	Erythroid TF	+0.231	$< 10^{-34}$
Gata2	Stem/progenitor TF	−0.204	$< 10^{-27}$

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Table N.36: Spinoids pseudotime–marker correlations (spinal cord organoid).

Gene	Function	Spearman ρ	p-value
MKI67	Proliferation (Ki-67)	+0.492	$< 10^{-183}$
TOP2A	Proliferation (topoisomerase)	+0.438	$< 10^{-141}$
NES	Neural progenitor (nestin)	+0.198	$< 10^{-28}$
TUBB3	Neuronal β-tubulin	−0.154	$< 10^{-17}$
SOX2	Neural progenitor	+0.136	$< 10^{-14}$
NEUROG1	Neurogenin-1	−0.076	$< 10^{-5}$

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